

~~An Interpretable~~ A Self-Explaining Temporal Graph Learning Method for ~~Real-Time~~ Low-Latency Financial Fraud Detection and Decision Making Support

Abstract

~~Financial fraud detection systems face a critical challenge: while Graph Neural Networks~~ Fraud screening systems must rank suspicious transactions quickly and tell the analyst why a transaction was flagged. Graph neural networks (GNNs) ~~have demonstrated strong predictive performance, they often rely on post hoc explanation techniques that introduce substantial computational overhead and limit their suitability for real-time, regulation-compliant decision support. This limitation is particularly important in environments subject to transparency requirements, such as the General Data Protection Regulation (GDPR) Article 22 and the Equal Credit Opportunity Act (ECOA).~~ To address this challenge, this study proposes RTXGNN, an intrinsically interpretable Graph Neural Network framework that jointly optimizes fraud detection performance and real-time explainability without requiring post hoc explanation generation. RTXGNN incorporates two key components: (1) Hierarchical Recency-Aware Positional Encoding exploit transaction relations but usually rely on post-hoc explainers, which are costly and may not reflect the model. We propose RTXGNN, a self-explaining temporal GNN whose learned feature, edge and node masks gate message passing and are returned as the explanation of every score. A training objective makes the top- k explanation sufficient for the prediction and its complement uninformative, and a shift-invariant temporal encoding (HRAPE) , ~~which captures multi-scale temporal characteristics including recency , periodicity, supplies time gaps, recency and transaction velocity; and (2) a Self-Explainable Aggregation Layer (SEAL), which generates multi-granularity explanations during the inference process. The proposed framework is evaluated on ,~~ On the Elliptic Bitcoin transaction dataset , which contains 203,769 transactions across 49 temporal steps. Experimental results demonstrate strong predictive performance,

achieving an F1-score of 0.5129 and an area under the receiver operating characteristic curve (AUC) of 0.8661, representing a 19.3% improvement in F1-score over the strongest baseline method. Ablation analyses show that HRAPE contributes a 2.53 percentage-point improvement in F1-score, while sparsity regularization yields an additional 6.37 percentage-point gain. Furthermore, RTXGNN preserves 76% of its full-supervision performance when trained with only 10% labeled data and achieves positive explanation fidelity (0.0764) dataset with a temporal split and 16 baselines over 10 seeds. RTXGNN reaches a test F1 of 0.700 and an average precision of 0.771, on par with the strongest GNN baselines and higher on average than the self-explaining SEFraud (+0.054 F1). The results suggest that explainability-oriented design can serve not only as a transparency mechanism but also as a form of domain-informed regularization, enhancing both predictive performance and interpretability in financial fraud detection.

Gradient-boosted trees on the same engineered features remain the most accurate model (F1 0.790). Its ranking reaches a precision of 0.99 among the 500 highest-scored test transactions. Its explanations are nearly as sufficient as Integrated Gradients at about one hundredth of the cost, and far more stable than GNNExplainer. On a synthetic benchmark with planted laundering rings, the most important incoming transaction of a detected ring member is a ring transaction in 91% of the cases. Scoring and explaining a transaction end to end takes 5.9 ms at the 99th percentile on one CPU core. We also analyse the temporal encoding, drift and retraining, label efficiency and a second financial dataset.

Keywords: Financial fraud analytics ; Explainable analytics ; Graph learning ; Decision support ; Temporal networks ; Regulatory compliance , Explainable analytics , Graph neural networks , Decision support , Temporal networks , Anti-money laundering

List of ~~Abbreviations~~ abbreviations

Table 1: Abbreviations used in this paper.

Abbreviation	Full term
<i>Regulatory and legal</i>	

Continued on next page

Abbreviation	Full term
AML/CFT	Anti-Money Laundering / Countering the Financing of Terrorism
ECOA	Equal Credit Opportunity Act
FATF	Financial Action Task Force
GDPR	General Data Protection Regulation
MiCA	Markets in Crypto-Assets Regulation
<i>Machine learning</i>	
AP	Average Precision (area under the precision–recall curve)
AUC	Area Under the Receiver Operating Characteristic Curve
Fid ⁺ / Fid [−]	Fidelity when the explanation is removed / kept alone
GCN, GAT	Graph Convolutional Network, Graph Attention Network
GNN	Graph Neural Network
GRU, LSTM	Gated Recurrent Unit, Long Short-Term Memory
IG	Integrated Gradients
KL	Kullback–Leibler divergence
LR, RF	Logistic Regression, Random Forest
MLP	Multi-Layer Perceptron
XAI	Explainable Artificial Intelligence
<i>Proposed method</i>	
RTXGNN	Real-Time eXplainable Graph Neural Network (the proposed method)
HRAPE	Hierarchical Recency-Aware Positional Encoding
SEAL	Self-Explainable Aggregation Layer
<i>Baselines</i>	
APAN	Asynchronous Propagation Attention Network
CARE-GNN	Camouflage-Resistant Graph Neural Network
EvolveGCN	Evolving Graph Convolutional Network
FRAUDRE	Fraud detection dual-resistant to graph inconsistency and imbalance
GAS	GCN-based Anti-Spam model
PC-GNN	Pick-and-Choose Graph Neural Network
SEFraud	Self-Explainable Fraud detection

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Abbreviation	Full term
TGAT, TGN	Temporal Graph Attention network, Temporal Graph Network

1. Introduction

The global cryptocurrency market, valued at over \$2.5 trillion as of 2024, has become a focal point for sophisticated financial crime, with illicit transactions estimated at \$Illicit activity in cryptocurrency markets was estimated at USD 24.2 billion annually in 2023¹. Unlike traditional banking systems with centralized oversight, blockchain networks present unique challenges for fraud detection: pseudonymous transactions, irreversible transfers, and complex networks of interrelated entities that fraudsters exploit to obfuscate illicit activities. As regulatory frameworks worldwide tighten—exemplified by the EU’s Blockchain transactions are pseudonymous and irreversible, and funds can be moved through long chains of intermediate addresses. Regulators have extended anti-money-laundering obligations to virtual asset service providers, for example through the FATF guidance on virtual assets and the EU Markets in Crypto-Assets (MiCA) regulation and the Financial Action Task Force’s guidance on virtual assets—financial institutions face mounting pressure to deploy fraud detection systems that are not only accurate but also transparent and explainable²³. Institutions are expected to flag suspicious transactions promptly and to document why a transaction was flagged.

Graph Neural Networks (GNNs) have emerged as the dominant paradigm for financial fraud detection, leveraging the relational structure inherent in transaction networks to achieve superior performance over traditional machine learning approaches. Building upon foundational architectures such as Graph Convolutional Networks (Kipf and Welling, 2017; Asiri and Somasundaram, 2025), Graph Attention Networks (Veličković et al., 2018; Fu et al., 2026), and

¹<https://go.chainalysis.com/crypto-crime-2024.html>

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²<https://www.fatf-gafi.org/en/publications/Fatfrecommendations/Guidance-rba-virtual-assets-2021.html>

³<https://www.esma.europa.eu/esmas-activities/digital-finance-and-innovation/markets-crypto-assets-regulation-mica>

GraphSAGE (Hamilton et al., 2017; Tang, 2025), specialized fraud detection methods have demonstrated substantial improvements. Recent are a natural model for this setting because they use the relations between transactions and accounts (Kipf and Welling, 2017; Velićković et al., 2018; Hamilton et al., 2017). Fraud-specific architectures such as CARE-GNN (Dou et al., 2020), which addresses camouflaged fraudsters through reinforcement learning-based neighbor selection, and PC-GNN (Liu et al., 2021), which tackles severe class imbalance through label-balanced sampling, routinely achieve AUC scores above 0.93 on benchmark datasets. However, this predictive power comes at a critical cost: *explainability*. The "black-box" nature of GNNs fundamentally conflicts with regulatory requirements such as GDPR Article 22, which mandates "meaningful information about the logic involved" in automated decisions, and the Equal Credit Opportunity Act (ECOA), which requires specific reasons for adverse actions (U.S. Congress, 1974).

The explainability challenge in GNN-based fraud detection manifests in three dimensions. *First*, existing post-hoc explanation methods address camouflage and class imbalance. Two practical obstacles remain for decision support. First, a flagged transaction is reviewed by an analyst, who needs to know which inputs drove the score. Post-hoc explainers such as GNNExplainer (Ying et al., 2019) and PGExplainer (Luo et al., 2020) require separate optimization procedures that consume 50-200ms per explanation, rendering them impractical for real-time fraud analysis processing millions of daily transactions (Lu et al., 2022). *Second*, these methods assume static graphs and cannot explain which temporal patterns triggered predictions—a critical gap given that fraudsters explicitly exploit temporal dynamics through velocity attacks, dormant account reactivation, and coordinated timing (Weber et al., 2019). *Third*, the field suffers from an "evaluation vacuum": existing evaluation methods for GNN explanations often rely on problematic ground-truth comparisons that can be fundamentally misleading (Faber et al., 2021), yet no standardized framework exists to assess what constitutes a "good" explanation in fraud investigation contexts provide this, but they require an additional optimisation or model per explanation and are not guaranteed to reflect the model's reasoning (Faber et al., 2021). Second, transaction networks change over time, and a model trained on past data can fail abruptly when the underlying activity changes (Weber et al., 2019; Gama et al., 2014). In addition, strong feature-based learners such as random forests are hard to beat on engineered transaction features (Weber et al., 2019; Grinsztajn et al., 2022). Claims of an advantage for a graph model therefore require careful comparison.

Recent research has begun addressing these limitations through self-explainable architectures that generate interpretations intrinsically during the prediction process. Self-explaining GNNs compute an explanation in the same forward pass as the prediction. SEFraud (Li et al., 2024), deployed at the Industrial and Commercial Bank of China (ICBC), represents a breakthrough by jointly learning prediction and interpretation through learnable feature and edge masks, achieving millisecond-level explanation generation while maintaining detection accuracy with 97% AUC and 0.4ms inference time. However, SEFraud operates on static graphs and lacks temporal awareness—a significant limitation given that financial fraud patterns evolve continuously. Temporal GNN architectures for general domains exist, for fraud detection on static heterogeneous graphs and reports industrial deployment. Temporal GNNs such as EvolveGCN (Pareja et al., 2020), which uses RNNs to evolve GNN parameters over time, and ROLAND (You et al., 2022), which enables incremental training for dynamic graphs. However, fraud-specific temporal explainability methods remain nascent, with current approaches unable to differentiate the importance of multiple events between node pairs based on timing. TGAT (Xu et al., 2020), TGN (Rossi et al., 2020) and APAN (Wang et al., 2021) model the evolution of graphs, but do not produce explanations.

Research Objective and Approach

This paper addresses a critical gap in fraud detection systems: the inability to simultaneously achieve high prediction accuracy, real-time inference, and regulatory-compliant explainability in temporal transaction networks. Current approaches face a fundamental trilemma—high-performing GNNs lack transparency; post-hoc explanation methods introduce prohibitive latency for production deployment; and existing self-explainable models ignore temporal dynamics essential for detecting evolving fraud patterns. Our objective is to develop an intrinsically interpretable GNN framework that jointly optimizes fraud detection accuracy and explanation quality in real-time while capturing multi-scale temporal patterns that characterize cryptocurrency fraud.

To achieve this objective, we introduce a novel framework that unifies temporal awareness with intrinsic explainability for cryptocurrency fraud detection. RTXGNN addresses the identified gaps through two core innovations: studies whether a self-explaining temporal GNN can provide explanations that are faithful to its predictions, at a latency compatible with online screening, while remaining competitive in detection accuracy with strong graph and non-graph baselines. We propose RTXGNN, which combines two components:

1. **Hierarchical Recency-Aware Positional Encoding (HRAPE):**

~~A temporal encoding scheme that captures~~, a temporal encoding built only from relative quantities: multi-scale dynamics by combining exponential decay functions for recency weighting with sinusoidal encodings for cyclical patterns (hour-of-day, day-of-week). Unlike simple timestamp concatenation or time-delta features used in prior temporal graph methods (Sankar et al., 2020), HRAPE provides a continuous, learnable representation that enables the model to distinguish fraud patterns that manifest at different temporal scales.

sinusoidal codes of the time gap between a transaction and the scored node, a learnable recency bias of the attention, and transaction counts in trailing windows (velocity). Because HRAPE never encodes the absolute timestamp, it does not extrapolate to unseen time values when the model scores future periods.

2. **~~An~~ Self-Explainable Aggregation Layers (SEAL)**, attention-based message-passing ~~mechanism that learns sparse, differentiable masks for~~ nodes, edges, and features during aggregation. By incorporating sparsity regularization into the training objective ~~layers whose messages are gated by learned edge and node masks~~, SEAL generates concise, interpretable explanations without requiring post-hoc optimization, achieving the real-time performance necessary for production deployment while avoiding the computational overhead of methods like GNNExplainer (Ying et al., 2019) ~~preceded by a per-node feature mask on the raw inputs. The masks that gate the computation are returned as the explanation.~~

The two are trained with an objective that makes the reported explanation faithful to the prediction. The top- k explanation alone must reproduce the prediction (sufficiency), and the inputs outside the explanation must not carry information about the label (necessity). A sparsity term keeps explanations short.

We evaluate RTXGNN on the Elliptic Bitcoin dataset (Weber et al., 2019) ~~with a temporal split (train on time steps 1–30, validate on 31–34, test on 35–49)~~, on the T-Finance transaction network (Tang et al., 2022), and on a synthetic benchmark in which laundering rings differ from decoy cycles only in the timing of their transactions. We compare with 16 baselines: feature-only models (logistic regression, random forest, XGBoost, MLP), general GNNs, temporal GNNs (EvolveGCN, TGAT, TGN, APAN) and fraud-specific GNNs (CARE-GNN, PC-GNN, GAS, FRAUDRE, SEFraud).

All results are averaged over 10 seeds with paired significance tests.

Summary of findings

On Elliptic, RTXGNN reaches a test F1 of 0.700 and an AP of 0.771. This is on par with the best graph neural network baseline, GAS (F1 0.701), and the differences to most other GNNs are within the variation across seeds. On average RTXGNN is ahead of the self-explaining SEFraud (+0.054 F1, +0.029 AP), although neither difference remains significant after correction for multiple comparisons. Gradient-boosted trees on the same engineered features remain the most accurate model (XGBoost, F1 0.790), and the random forest reaches 0.733. For decision support the ranking is what matters: the 500 highest-scored test transactions have a precision of 0.99 (Section 5.2). Reading the explanation from the masks adds 0.24 ms to a widely-adopted benchmark consisting of 203,769 transactions and 234,355 payment flows labeled as licit, illicit, or unknown. Our temporal evaluation protocol follows realistic deployment scenarios: training on time steps 1-30, validating on steps 31-34, and testing on future steps 35-49 to assess generalization under concept drift. The experimental design includes comprehensive ablation studies to isolate the contributions of HRAPE and SEAL, label efficiency analysis to evaluate semi-supervised learning capabilities, and temporal stability assessments to quantify robustness to evolving fraud patterns. Single-transaction request. The explanations' sufficiency is close to that of Integrated Gradients, which is about one hundred times more expensive, and they are far more stable than GNNExplainer (Section 5.7). On a synthetic benchmark with planted laundering rings, the most important transaction identified by RTXGNN is a ring transaction for 91% of the detected ring members (Section 5.8). All models fail after the dark-market shutdown in the Elliptic test period, and retraining with a one-step label delay recovers part of the loss (Section 5.5). End-to-end scoring and explanation of a single transaction, including neighbourhood sampling, takes 5.9 ms at the 99th percentile on one CPU core (Section 5.10).

Contributions

This work makes three principal contributions:

1. We propose RTXGNN, the first self-explainable GNN framework that jointly models temporal dynamics and generates multi-granularity explanations (node, edge, feature, temporal) during inference without post-hoc computation. HRAPE captures multi-scale temporal patterns—recency,

periodicity, and velocity—that static models miss, while SEAL intrinsically generates sparse interpretable masks during message-passing, achieving sub-millisecond explanation generation suitable for real-time deployment.

2. ~~÷ We demonstrate that sparsity-regularized intrinsic explainability improves rather than degrades detection performance, with SEAL’s sparse masking contributing +6.37pp F1 and HRAPE contributing +2.51pp F1. This challenges the conventional explainability-performance trade-off, showing that well-designed interpretability constraints act as domain-informed feature selection that reduces overfitting.~~

A self-explaining temporal GNN with a faithfulness objective.

RTXGNN gates message passing with learned feature, edge and node masks, and it is trained with explicit sufficiency and necessity terms. These terms rule out masks that are not used by the model. HRAPE uses only shift-invariant temporal information. In our ablation it is more accurate than Time2Vec on the same time gaps, and adding the absolute timestamp makes performance less stable across seeds under a temporal split.

3. ~~÷ We establish evaluation protocols addressing gaps identified in recent surveys (Yuan et al., 2023; Cherif et al., 2024; ?): temporal stability analysis across 15 future time steps; label efficiency assessment showing 76% performance retention with only~~ **A reproducible, statistically controlled comparison.** We compare 16 baselines, including tree ensembles and temporal GNNs, under one protocol with 10 % labeled data; explanation fidelity metrics that avoid problematic ground-truth comparisons (Faber et al., 2021); and case studies demonstrating explanations actionable under GDPR Article 22 and ECOA (U.S. Congress, 1974)–requirements ~~seeds and Holm-corrected paired tests, and we release all code. The evaluation includes ablations that replace HRAPE with Time2Vec, a comparison of mask sparsity with dropout and weight decay, stratified label-efficiency experiments with repeated subsets, a per-time-step drift analysis with a retraining strategy, and measured end-to-end CPU latency with tail percentiles.~~
4. **An evaluation of explanations for decision support.** We measure the sufficiency and necessity of feature explanations at several sizes against saliency, Integrated Gradients, GNNExplainer, SEFraud’s masks and random explanations, together with class-wise results, stability and cost. We test edge explanations against known ground truth on

~~synthetic data, translate detection scores into alert-queue terms, and discuss what is needed to use such explanations in regulated review processes.~~

~~The remainder of this paper is organized as follows. Section~~Section 2 ~~re-~~
~~views related work on GNN-based fraud detection, temporal graph learning,~~
~~and explainable AI in financial contexts. Section~~3 ~~formalizes the problem~~
~~and presents the RTXGNN architecture in detail. Section~~5 ~~describes the~~
~~experimental setup, evaluation metrics, and comprehensive results including~~
~~ablation studies, label efficiency analysis, temporal stability assessment, and~~
~~interpretable case studies. Section~~6 ~~discusses implications for regulatory~~
~~compliance, deployment considerations, and limitations. Section~~7 ~~concludes~~
~~with directions for future research. Section 3 presents RTXGNN. Section 4~~
~~describes the datasets, baselines and protocol, and Section 5 the results.~~
~~Section 6 discusses decision support, regulation, deployment and limitations,~~
~~and Section 7 concludes.~~

2. Related work

~~We organize the related work into five key areas: foundational graph~~
~~neural network architectures, GNN-based fraud detection methods, explainability~~
~~techniques for GNNs review graph neural networks for fraud detection, explainability~~
~~methods for GNNs and how they are evaluated, temporal graph learning, and~~
~~real-time deploymentsystems. This review positions RTXGNN’s contributions~~
~~within the existing research landscape and identifies the gaps our work addresses~~low-latency
~~deployment. We then position RTXGNN against the closest methods.~~

2.1. Graph neural networks for fraud detection

~~Graph Neural Networks have become the dominant paradigm for fraud~~
~~detection due to their ability to model relational dependencies in~~model the
relational structure of ~~transaction networks. The foundational architectures~~
~~—Graph Convolutional Networks (GCN) (Kipf and Welling, 2017; Asiri~~
~~and Somasundaram, 2025), Graph Attention Networks (GAT) (Veličković~~
~~et al., 2018; Fu et al., 2026) —and GraphSAGE (Hamilton et al., 2017; Tang,~~
~~2025) —established the core~~established the ~~message-passing framework that~~
~~subsequent fraud detection methods build upon. GCN introduced spectral~~
~~graph convolutions with linear complexity in graph edges, GAT incorporated~~
~~attention mechanisms for weighted neighborhood aggregation on which most~~

fraud detectors build. GCN aggregates neighbours with a normalised adjacency, GAT learns attention weights over neighbours, and GraphSAGE enabled inductive learning that generalizes learns inductive aggregators that generalise to unseen nodes—critical for production fraud systems encountering novel accounts, which matters when new accounts or transactions arrive continuously.

However, direct application of these general-purpose architectures to fraud detection faces significant challenges, where fraudsters deliberately Fraud graphs violate several assumptions of these architectures. Camouflage, in which fraudsters mimic legitimate users, undermines standard neighborhood aggregation assumptions weakens neighbourhood aggregation. CARE-GNN (Dou et al., 2020) addresses this through reinforcement learning-based neighbor selection, using selects neighbours with a label-aware similarity measures to distinguish genuine behavioral patterns from camouflaged activities. The method demonstrates substantial improvements by explicitly modeling both feature camouflage (fraudsters adopting legitimate user characteristics) and relation camouflage (fraudsters establishing connections with normal users).

measure and a reinforcement-learning threshold. **Class imbalance** presents another fundamental challenge—fraudulent transactions typically constitute less than 1% of total volume. is addressed by PC-GNN (Liu et al., 2021) tackles this through a "pick-and-choose" sampling strategy that balances label distribution while selecting informative neighbors, achieving 3-5% AUC improvements on benchmark datasets. The method combines label-balanced sub-graph sampling with a picking function that identifies neighbors most relevant for classification.

arise when applying GNNs to fraud graphs, where fraudulent and benign nodes exhibit different neighborhood patterns. Liu et al. (Liu et al., 2020) identify three types of inconsistencies: context inconsistency (same features yield different labels depending on neighbors), feature inconsistency (fraudsters share high feature similarity with legitimate users), and relation inconsistency (edge semantics vary between fraud and non-fraud contexts). Their GraphConsis framework addresses these through context embeddings and consistency-based filtering.

Recent architectures have focused on label-balanced sampling of training nodes and neighbour selection. FRAUDRE (Zhang et al., 2021) tackles both graph inconsistency and imbalance with a fraud-aware convolution that uses node-neighbour differences, a combination of intermediate layer outputs, and an imbalance-oriented classifier. GAS (Li et al., 2019), deployed at Xianyu,

combines aggregation over the interaction graph with aggregation over a similarity graph built from content features. Liu et al. (2020) characterise context, feature and relation inconsistencies and propose GraphConsis. H2-FDetector (Shi et al., 2022) recognizes that fraud graphs contain both homophilic edges (similar nodes connected) and heterophilic edges (dissimilar nodes connected), requiring differentiated aggregation strategies. GeniePath (Liu et al., 2019), deployed in production at Alipay, introduces adaptive path layers that learn application-specific receptive fields through breadth and depth exploration, separates homophilic from heterophilic edges, GeniePath (Liu et al., 2019) learns adaptive receptive fields, and DGA-GNN (Duan et al., 2024) addresses non-additivity of node attributes through decision tree binning and dynamic grouping, achieving 3-16% improvements over existing methods.

While these fraud-specific GNNs advance detection accuracy, they largely operate as black boxes, providing predictions without explanations—a critical gap RTXGNN addresses through intrinsic explainability groups non-additive attributes with decision-tree binning. Tang et al. (2022) relate fraud and anomaly detection on graphs to spectral “right-shift” effects and release the T-Finance dataset used in our study. GADBench (Tang et al., 2023) shows that tree ensembles with neighbourhood aggregation are strong competitors to specialised GNNs on many anomaly-detection benchmarks.

On the Elliptic Bitcoin data specifically, Weber et al. (2019) found that a random forest on the transaction features outperforms GCN, and that all models fail after a dark-market shutdown in the test period. This is consistent with the broader observation that tree ensembles remain very strong on tabular features (Grinsztajn et al., 2022). We therefore include logistic regression, random forest (Breiman, 2001) and XGBoost (Chen and Guestrin, 2016) as baselines, in addition to general, temporal and fraud-specific GNNs.

Explainability techniques for GNNs fall into two categories: post-hoc methods that explain predictions after model training, and intrinsic methods that generate explanations as part of the prediction process.

2.2. Explainability methods for graph neural networks

typically identify important subgraphs and features through optimization or perturbation. GNNExplainer (Ying et al., 2019), the foundational work in this area, formulates explanation as maximizing mutual information between the prediction and a candidate subgraph structure. While model-agnostic and theoretically grounded, GNNExplainer requires separate optimization for each prediction, consuming 50-200ms per explanation—prohibitive for

~~real-time fraud systems processing millions of transactions~~ **Post-hoc explainers** explain a trained model. GNNExplainer (Ying et al., 2019) optimises soft masks over edges and features that preserve a prediction and must be run separately for every instance. PGExplainer (Luo et al., 2020) ~~addresses this computational bottleneck through a parameterized approach using deep neural networks to generate edge importance scores collectively across instances, achieving up to 24.7% AUC improvement in explanation quality while providing inductive capability.~~

SubgraphX (Yuan et al., 2021) ~~advances explanation quality through explicit subgraph identification using Monte Carlo Tree Search combined with Shapley values to measure importance while capturing interaction effects between nodes. However, its computational requirements make it suitable only for offline analysis rather than real-time deployment~~ amortises this optimisation with a parametric edge-mask network. SubgraphX (Yuan et al., 2021) searches subgraphs with Monte Carlo tree search and Shapley values and is accurate but slow. CF-GNNExplainer (Lucic et al., 2022) ~~provides counterfactual explanations by identifying minimal edge perturbations that would change predictions, achieving 94%+ accuracy while removing fewer than 3 edges on average. Such counterfactual explanations offer theoretical advantages for regulatory compliance by answering "what would need to change to flip this decision?" but the method supports only edge deletions, not the additions or feature changes relevant to fraud scenarios.~~

~~Alternative post-hoc approaches include GraphLIME (Huang et al., 2023) , which extends LIME to graphs using HSIC Lasso for nonlinear feature importance, and XGNN (Yuan et al., 2020) , which provides~~ finds minimal edge deletions that change a prediction, and CF² (Tan et al., 2022) combines factual (sufficient) and counterfactual (necessary) criteria. GraphLIME (Huang et al., 2023) fits local surrogate models, XGNN (Yuan et al., 2020) gives model-level rather than instance-level explanation ~~through reinforcement learning. Yuan et al.'s comprehensive survey (Yuan et al., 2023) provides a unified taxonomy covering gradient-based explanations, perturbation-based, decomposition, and surrogate approaches, but notes that most methods remain evaluated primarily on synthetic benchmarks rather than real-world fraud datasets.~~

~~for GNN explanations is particularly acute. Faber et al. (Faber et al., 2021) demonstrate that ground-truth-based evaluation is fundamentally problematic, identifying five critical pitfalls: redundant explanations achieving high scores, weak predictors being rewarded, GNNs using different rationale than ground truth, trivial baselines succeeding, and gradient saturation effects. This~~

finding is consequential—most explainability papers evaluate against synthetic ground truth that may not reflect actual model reasoning. Agarwal et al.’s GraphXAI framework (Agarwal et al., 2023) addresses this through synthetic data generation with known explanatory motifs, but such synthetic benchmarks do not replicate the complex patterns in real fraud networks. The field lacks fraud-specific evaluation protocols, particularly for assessing whether explanations actually aid human fraud analysts in decision-making and gradient-based attributions such as saliency (Simonyan et al., 2014) and Integrated Gradients (Sundararajan et al., 2017) apply to GNNs directly (Pope et al., 2019; Baldassarre and A. Yuan et al. (2023) survey the field.

offers an alternative paradigm. Rather than explaining predictions post-hoc, self-explainable models generate interpretations as part of the prediction process. **Self-explaining GNNs** compute explanations during the forward pass. GSAT (Miao et al., 2022) learns stochastic attention under an information bottleneck. SEFraud (Li et al., 2024) represents the breakthrough work in this direction—learns feature and edge masks for fraud detection, using heterogeneous graph transformers with learnable feature and edge masks. Deployed on heterogeneous graphs with a triplet loss that separates the masked from the complementary representation, and reports deployment at the Industrial and Commercial Bank of China, SEFraud achieves 1000× speedup over GNNExplainer for explanation generation while maintaining 97% AUC with 0.4ms inference time. However, SEFraud operates. SEFraud and GSAT operate on static graphs and provides no temporal explanations. RTXGNN extends this self-explainable paradigm to temporal graphs, addressing both the computational efficiency demonstrated by SEFraud and the temporal reasoning gap left unaddressed. RTXGNN adopts the self-explaining design but uses an objective that ties the reported top- k explanation to the prediction through explicit sufficiency and necessity terms, and it adds relative temporal encodings. We use the term “self-explaining” rather than “interpretable”: a network with several non-linear layers remains a complex model even when it exposes attributions (Rudin, 2019; Lipton, 2018).

Most GNN fraud detection methods treat transaction graphs as static snapshots, ignoring the critical temporal dimension. Fraud patterns evolve continuously—attackers adapt strategies, dormant accounts reactivate, and coordinated attacks exhibit timing patterns. Temporal GNNs for general dynamic graphs have been extensively studied, but fraud-specific temporal methods remain limited. **Evaluating explanations** is difficult. Faber et al. (2021) show that comparing explanations with synthetic ground truth can mislead,

because a model may solve a task with a rationale different from the planted one. Fidelity metrics measure instead how the prediction changes when the explanation is removed (Fid^+ , necessity) or kept alone (Fid^- , sufficiency) (Yuan et al., 2021; Amara et al., 2022). Agarwal et al. (2023) add stability and fairness criteria. We report Fid^+ and Fid^- at several explanation sizes, with a random explanation as reference. We also report stability under input perturbation and agreement across independently trained models, class-wise results, and the cost of each explainer. Ground truth is used only on the synthetic benchmark, where the planted rings are known to be the only signal separating the classes.

~~EvolveGCN (Pareja et al., 2020) adapts graph convolutions to temporal dynamics by using RNNs (GRU/LSTM) to evolve weight parameters over time rather than maintaining node embeddings. This approach offers EvolveGCN-H (evolving convolution weights) and~~

2.3. Temporal graph neural networks

~~Discrete-time models operate on snapshots. EvolveGCN (Pareja et al., 2020) evolves the GCN weights with a recurrent network (EvolveGCN-O (evolving graph normalization) variants. Rossi et al.’s Temporal Graph Networks (Rossi et al., 2020) provide a generic framework for continuous-time dynamic graphs using memory modules, demonstrating that previous methods (/-H), DySAT (Sankar et al., 2020) applies structural and temporal self-attention, and ROLAND (You et al., 2022) turns static GNNs into incremental dynamic models. Continuous-time models consume timestamped events. TGAT (Xu et al., 2020) encodes time gaps with a functional time encoding inside temporal attention. TGN (Rossi et al., 2020) adds a per-node memory updated by a recurrent cell from event messages and shows that JODIE, DyRep , TGAT) are specific instances of this unified architecture. and TGAT are special cases. APAN (Wang et al., 2021) decouples inference from graph queries with asynchronous mailboxes, a design aimed at real-time serving. Time2Vec (Kazemi et al., 2019) is a generic learnable time embedding with one linear and several periodic components.~~

~~For fraud detection specifically, temporal modeling is critical but underexplored. The Elliptic Bitcoin dataset (Weber et al., 2019), the primary benchmark for cryptocurrency fraud detection, contains temporal information across Continuous-time models assume that nodes interact repeatedly over time. In Elliptic each transaction exists in one of 49 time steps, yet most methods ignore this dimension. ROLAND (You et al., 2022) offers a promising framework~~

for adapting any static GNN to dynamic graphs through hierarchical node states with recurrent updates, introducing live-update evaluation and scalable incremental training—both essential for production fraud systems. DySAT (Sankar et al., 2020) employs self-attention along both structural neighborhood and temporal dimensions for capturing evolutionary patterns.

However, . Current methods cannot explain which temporal patterns triggered predictions, why specific time windows matter, or how fraud network evolution influenced decisions. RTXGNN’s HRAPE mechanism explicitly captures multi-scale temporal dynamics through recency and periodicity encoding, while SEAL provides explanations that identify not just important nodes and features but also their temporal relevancesnapshots and all edges connect transactions of the same snapshot (Weber et al., 2019). The memory of TGN and the mailbox of APAN therefore carry no information across snapshots, and time gaps within a snapshot are zero. We nevertheless include TGAT, TGN, APAN and EvolveGCN as baselines, because they are the standard temporal GNNs, and describe their adaptation in Appendix B. DySAT’s temporal attention runs over the snapshots in which a node appears. Every Elliptic transaction appears in exactly one snapshot, so DySAT reduces to a structural attention network (GAT), which is already a baseline.

2.4. Low-latency fraud detection systems

Production fraud detection systems face stringent latency requirements—typically sub-10ms to 100ms have end-to-end —creating fundamental tension between model complexity and deployment feasibilitylatency budgets of roughly 10–100 ms. BRIGHT (Lu et al., 2022), deployed at eBay/PayPal, addresses this through Lambda architecture that , pre-computes entity embeddings in batch layers while reserving real-time inference for new transaction links, achieving $7.8\times$ speedup with $>75\%$ P99 latency reduction and $>2\%$ precision improvement over baseline systems.

TitAnt (Cao et al., 2019; Dong and Xiao, 2025) , Ant Financial’s production system, demonstrates millisecond-level fraud prediction for large-scale transaction data through sophisticated feature engineering and model optimization. GEM (Liu et al., 2018; Rao et al., 2025) , deployed at Alipay for malicious account detection, achieves $>98\%$ precision in production while covering 10% more accounts than rule-based methods, demonstrating that heterogeneous GNNs can meet production requirements.

These production systems highlight a common pattern: . BRIGHT uses pre-computed embeddings, eliminating the possibility of real-time explanation

generation. ~~TitAnt and GEM provide prediction scores but no interpretations. The dominant production pattern is tiered explainability: lightweight mechanisms for real-time decisions, comprehensive explanations generated asynchronously for audit trails. RTXGNN demonstrates that this trade-off is not fundamental—intrinsic explainability through SEAL achieves sub-millisecond explanation-generation without sacrificing prediction accuracy or a batch layer and runs only a light real-time performance layer for new transactions. TitAnt (Cao et al., 2019; Dong and Xiao, 2025) and GEM (Liu et al., 2018; Rao et al., 2025) serve predictions at Ant Financial and Alipay. These systems return scores without explanations. When explanations are needed, they are usually produced asynchronously for the cases that are escalated. Latency figures reported in the literature depend strongly on hardware, batch size and whether neighbourhood retrieval is included. We therefore report our own end-to-end measurements on commodity CPU hardware, with every stage of the pipeline timed separately.~~

Beyond ~~cryptocurrency fraud, supervised machine learning and deep learning approaches have demonstrated strong performance across financial fraud domains more broadly. Afriyie et al. (2023) conducted a systematic comparison of graph data, supervised learning is widely used for credit-card and insurance fraud. Afriyie et al. (2023) compare logistic regression, decision tree, and random forest classifiers for credit-card fraud detection, establishing that random forest achieves superior AUC (0.989) and accuracy (96%) under severe class imbalance—findings that motivate our adoption of focal loss and class-weighted training as complementary imbalance-mitigation strategies. In the insurance domain, Jacob et al. (2025) proposed a hybrid trees and random forests for credit-card fraud and find the random forest strongest under class imbalance. Jacob et al. (2025) combine optimisation-based feature selection with a BERT-LSTM architecture preceded by Golden Eagle-assisted feature selection, achieving 99.02% accuracy on the careclaims benchmark; this confirms that intrinsic feature selection—whether optimisation-based or, as in RTXGNN, sparsity-regularised masking—is a decisive factor in fraud detection performance. Unlike both methods, which treat input graphs or tabular features as static, RTXGNN explicitly models the temporal evolution of transaction networks and generates real-time, multi-granularity explanations aligned with regulatory transparency requirements model for insurance claims. These results reinforce the need to compare graph models against strong feature-based learners.~~

2.5. Regulation and explanation requirements

Regulatory frameworks increasingly demand explainable AI in financial services. ~~GDPR Article 22⁴ prohibits~~ GDPR Article 22⁴ restricts solely automated decisions with legal or similarly significant effects, ~~requiring~~ and Articles 13–15 require “meaningful information about the logic involved” ~~for affected individuals. The~~. Whether this amounts to a right to an explanation is debated (Wachter et al., 2017; Selbst and Powles, 2017; Goodman and Flaxman, 2017). In U.S. consumer credit, the Equal Credit Opportunity Act (U.S. Congress, 1974) ~~implemented through Regulation B, requires creditors to provide specific and accurate~~ and Regulation B require specific principal reasons for adverse actions. The ~~Consumer Financial Protection Bureau’s 2022 circular⁵~~ explicitly states that ECOA does not permit creditors to use complex algorithms when doing so prevents providing specific reasons—a direct challenge to black-box GNN deployments.

The academic debate on whether GDPR creates enforceable explanation requirements remains contentious. ~~Wachter et al. (Wachter et al., 2017)~~ argue that no explicit right to explanation exists in GDPR’s text, while Selbst and Powles (Selbst and Powles, 2017) counter that Articles 13–15 require meaningful information that effectively mandates explanations. Goodman and Flaxman (Goodman and Flaxman, 2016) examine the practical implications for algorithmic decision-making systems. The discussion has been continued recently (Sargeant, 2026).

~~For cryptocurrency specifically~~ CFPB has stated that this obligation also applies when complex algorithms are used⁵. For virtual assets, FATF guidance on virtual assets⁶ requires virtual asset service providers to implement the same AML and the EU MiCA regulation (Teng et al., 2026) require AML/CFT measures as financial institutions, creating audit trail requirements for AI systems flagging suspicious transactions. The EU’s Markets in Crypto-Assets regulation (Teng et al., 2026) establishes comprehensive oversight requiring crypto-asset service providers to implement governance and fraud prevention

⁴<https://eur-lex.europa.eu/eli/reg/2016/679/oj>

⁵

⁵<https://www.consumerfinance.gov/compliance/circulars/circular-2022-03-adverse-action-notification-requirements-in-connection-with-credit-decisions-based-on-complex-algorithms/>

⁶<https://www.fatf-gafi.org/en/publications/Fatfrecommendations/Guidance-rba-virtual-assets-2021.html>

systems with transparency requirements that implicitly demand explainability⁷. CFT controls that are documented and auditable. Supervisory guidance on model risk management, such as the U.S. SR 11-7 (Board of Governors of the Federal Reserve System), requires that models be validated and their limitations understood.

Despite this regulatory pressure, Cherif et al.’s systematic review (Cherif et al., 2024) identifies this as a critical gap—the field lacks standardized evaluation frameworks for fraud-specific explanation quality, regulatory compliance validation, or human-centered assessment with actual fraud analysts. These texts set requirements for processes and for the content of notices, not for model architectures. A feature attribution can contribute to a reason statement or an audit record, but it does not by itself establish compliance with any of them. This applies especially when the features are anonymised, as in Elliptic. Section 6.2 discusses what would be needed to operationalise RTXGNN’s explanations. Cherif et al. (2024) note that GNN explanations have not yet been validated with analysts or regulators, and our work does not close this gap.

2.6. Positioning of RTXGNN

RTXGNN addresses multiple identified gaps simultaneously. Unlike post-hoc explanation methods (Luo et al., 2020; Yuan et al., 2021), RTXGNN generates explanations intrinsically through SEAL, achieving Table 2 compares RTXGNN with the closest methods on properties that can be verified from the methods themselves. We do not include a “regulatory compliance” property, because compliance is a property of an institution’s process and cannot be verified from a method, and we report “explanation in the forward pass” rather than “real-time performance comparable to production systems (Lu et al., 2022; ?)”. Unlike SEFraud (Li et al., 2024), which operates on static graphs, RTXGNN incorporates temporal awareness through HRAPE, addressing the critical gap in temporal explainability. Unlike existing temporal GNNs (Pareja et al., 2020; You et al., 2022; Sa, which focus on prediction accuracy, RTXGNN jointly optimizes prediction and interpretation quality.

Our evaluation follows rigorous protocols addressing the ground-truth pitfalls identified by Faber et al. (Faber et al., 2021), using fidelity metrics from the GraphXAI framework (Agarwal et al., 2023) while providing human-interpretable case studies. By demonstrating that self-explainable architectures can achieve

⁷

Table 2: Properties of related methods. **F** = supported, *P* = partially supported, $-$ = not supported. Temporal: uses timestamps or snapshots. Self-expl.: explanation computed in the forward pass. Fraud: design for camouflage or imbalance. Feat./Edge: explanation over input features / edges. Obj.: explanation tied to the prediction by a sufficiency/necessity objective.

Method	Temporal	Self-expl.	Fraud	Feat.	Edge	Obj.
GCN / GraphSAGE (Kipf and Welling, 2017; Hamilton et al., 2017)	$-$	$-$	$-$	$-$	$-$	$-$
GAT (Velićković et al., 2018)	$-$	<i>P</i>	$-$	$-$	<i>P</i>	$-$
CARE-GNN / PC-GNN (Dou et al., 2020; Liu et al., 2021)	$-$	$-$	F	$-$	$-$	$-$
FRAUDRE / GAS (Zhang et al., 2021; Li et al., 2019)	$-$	$-$	F	$-$	$-$	$-$
EvolveGCN (Pareja et al., 2020)	F	$-$	$-$	$-$	$-$	$-$
TGAT / TGN / APAN (Xu et al., 2020; Rossi et al., 2020; Wang et al., 2021)	F	<i>P</i>	$-$	$-$	<i>P</i>	$-$
GNNE explainer (Ying et al., 2019) (post hoc)	$-$	$-$	$-$	F	F	<i>P</i>
PGExplainer (Luo et al., 2020) (post hoc)	$-$	$-$	$-$	$-$	F	<i>P</i>
GSAT (Miao et al., 2022)	$-$	F	$-$	$-$	F	<i>P</i>
SEFraud (Li et al., 2024)	$-$	F	F	F	F	<i>P</i>
RTXGNN (ours)	<i>P</i>	F	<i>P</i>	F	F	F

RTXGNN’s temporal support is marked partial because it uses relative time only, which is uninformative on Elliptic (Section 3.2). Its fraud-specific support is partial because it relies on class-balanced training rather than on a camouflage-specific mechanism. Attention weights (GAT, TGAT) are marked partial because attention is not necessarily faithful to the prediction.

~~production-grade performance without sacrificing interpretability, RTXGNN offers a path toward regulatory-compliant GNN deployment in high-stakes financial applications.~~

~~explanation”, since latency depends on the deployment.~~

~~Table 2 presents a systematic comparison of RTXGNN against closely related methods across six key dimensions. refers to explicit encoding of time-dependent patterns (recency, periodicity, velocity). indicates whether explanations are generated during inference without post-hoc optimization. denotes sub-10ms latency suitable for production deployment. indicates design tailored to fraud detection challenges (camouflage, imbalance, adversarial behavior). refers to identifying important nodes, edges, features, and temporal patterns simultaneously. indicates alignment with GDPR Article 22 and ECOA requirements for meaningful, actionable explanations.~~

The comparison reveals that existing approaches address subsets of requirements but no single method satisfies all dimensions. General-purpose GNNs (GCN, GAT, GraphSAGE) lack both temporal modeling and explainability. Fraud-specific GNNs (CARE-GNN, PC-GNN) address domain challenges but provide no explanations. Post-hoc explainers (GNNE explainer, PGExplainer, SubgraphX) operate on static graphs with prohibitive latency. Temporal GNNs (EvolveGCN, DySAT, ROLAND) capture dynamics but offer no interpretations. SEFraud

achieves intrinsic explainability for fraud detection but ignores temporal patterns critical for cryptocurrency transactions. RTXGNN uniquely combines temporal awareness (HRAPE) with intrinsic multi-granularity explainability (SEAL), providing the first framework suitable for regulatory-compliant real-time fraud detection in evolving transaction networks.

3. Methodology

This section presents the RTXGNN (Real-Time Explainable Graph Neural Network) framework, a novel architecture that jointly optimizes fraud detection accuracy and explanation quality through intrinsic interpretability mechanisms. We first formalize the problem (Section 3.1), then detail our two core innovations: and describes the two components of RTXGNN: the Hierarchical Recency-Aware Positional Encoding (HRAPE, Section 3.2) and the Self-Explainable Aggregation Layer (SEAL, Section 3.3). Finally, we describe the training procedure with multi-task learning objectives (Section 3.4). The architecture of RTXGNN is presented in Figure 1. Section 3.4 presents the training objective and Section 3.5 the inference procedure used for low-latency scoring. Figure 1 gives an overview. Every equation in this section corresponds one-to-one to the released implementation (module `rtxgnn/model.py` for the architecture and `rtxgnn/objective.py` for the objective). In this figure, the framework processes temporal transaction graphs through three main stages: (1) Hierarchical Recency-Aware Positional Encoding (HRAPE) captures multi-scale temporal patterns through recency decay, periodicity, and velocity features; (2) Self-Explainable Aggregation Layers (SEAL) generate sparse differentiable masks for features, edges, and nodes while performing message passing across $L=3$ layers; (3) Dual-head output produces both fraud predictions ($\hat{y}$) and multi-granularity explanations (M^{feat} , M^{edge} , M^{node} , M^{temp}) simultaneously without post-hoc optimization.

Terminology. Following Rudin (2019) and Lipton (2018), we distinguish *interpretable* models, whose decision logic can be inspected directly (e.g. sparse linear models, shallow decision trees), from *self-explaining* models, which remain complex non-linear predictors but compute an explanation as part of the forward pass. RTXGNN belongs to the second category. Its masks are attributions: they state which inputs the prediction depended on, not the full decision logic. We therefore do not describe it as intrinsically interpretable.

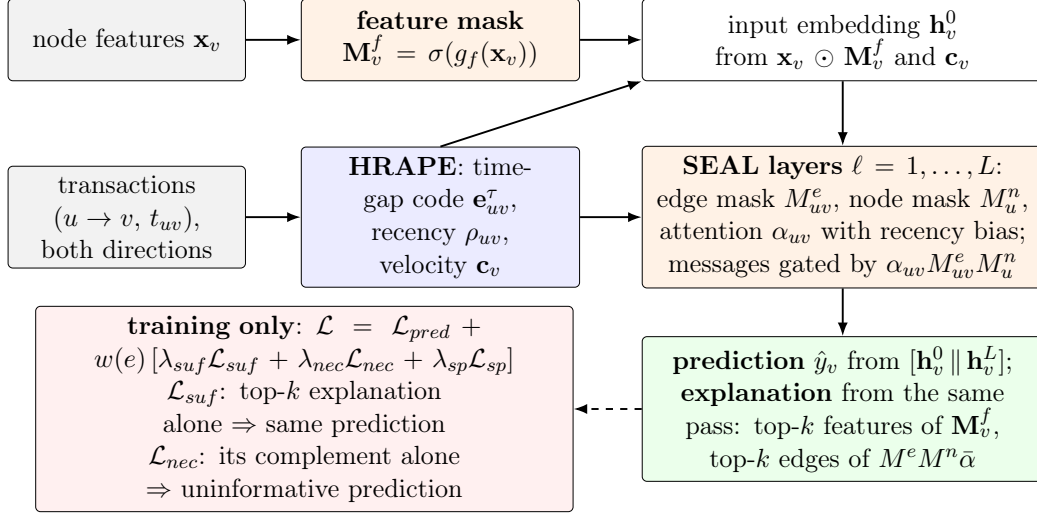


Figure 1: Overview of RTXGNN. The masks that form the explanation are the gates that produce the prediction, so both come out of one forward pass. The objective in the bottom box is used only during training. It evaluates the model twice more, once on the top- k explanation and once on its complement (Section 3.4).

3.1. Problem formulation

We model the financial transaction network as a dynamic temporal graph $\mathcal{G}^t = (\mathcal{V}, \mathcal{E}^t, \mathbf{X}^t, \mathbf{T}^t)$ at time t , where:

- $\mathcal{V} = \{v_1, \dots, v_n\}$ is the set of n entities (accounts, addresses, wallets);
- $\mathcal{E}^t \subseteq \mathcal{V} \times \mathcal{V}$ is the set of directed edges representing transactions at time t ;
- $\mathbf{X}^t \in \mathbb{R}^{n \times d_v}$ is the node feature matrix, where each row $\mathbf{x}_i \in \mathbb{R}^{d_v}$ contains behavioral and structural features for node v_i ;
- $\mathbf{T}^t \in \mathbb{R}^{|\mathcal{E}^t|}$ contains timestamps for each edge in $\mathcal{E}^t$.

Given a target node v_i (representing a transaction or account), predict the probability that v_i is involved in fraudulent activity:

$$\hat{y}_i = f_\theta(\mathcal{G}^t, v_i) \in [0, 1].$$

where θ denotes the model parameters.

Simultaneously generate a multi-granularity explanation $\mathcal{X}_i = \{\mathbf{M}_i^{\text{node}}, \mathbf{M}_i^{\text{edge}}, \mathbf{M}_i^{\text{feat}}, \mathbf{M}_i^{\text{temp}}\}$ comprising: A transaction network is a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each node $v \in \mathcal{V}$ (a transaction in Elliptic, an account in T-Finance and in the synthetic benchmark) has a feature vector $\mathbf{x}_v \in \mathbb{R}^d$ and a timestamp τ_v , the time at which it has to be scored. Each edge $(u \rightarrow v) \in \mathcal{E}$ is a flow of funds with timestamp $t_{uv} \leq \tau_v$. Message passing uses both directions of every edge. The reversed copy $(v \rightarrow u)$ carries the type $r = 1$ and the original the type $r = 0$, so the model can distinguish incoming from outgoing funds. The time gap of a directed edge is $\Delta_{uv} = |\tau_v - t_{uv}|$.

- $\mathbf{M}^{\text{node}} \in [0, 1]^n$: Node importance scores indicating which neighbors contribute to the prediction;
- $\mathbf{M}^{\text{edge}} \in [0, 1]^{|\mathcal{E}^t|}$: Edge importance scores identifying critical transaction paths;
- $\mathbf{M}^{\text{feat}} \in [0, 1]^{d_v}$: Feature importance scores revealing which attributes drive the decision;
- $\mathbf{M}^{\text{temp}} \in [0, 1]^{|\mathcal{E}^t|}$: Temporal importance scores capturing timing-dependent patterns.

The joint optimization objective is :-

$$\min_{\theta} \mathcal{L}_{\text{pred}}(\hat{y}, y) + \lambda_{\text{exp}} \mathcal{L}_{\text{exp}}(\mathcal{X}) + \lambda_{\text{sparse}} \mathcal{L}_{\text{sparse}}(\mathcal{X})$$

where $\mathcal{L}_{\text{pred}}$ measures prediction accuracy, $\mathcal{L}_{\text{exp}}$ ensures explanation fidelity, and $\mathcal{L}_{\text{sparse}}$ promotes concise explanations through sparsity regularization. Hyperparameters λ_{exp} and λ_{sparse} balance the three objectives. Given the labelled nodes observed up to a cut-off time, the task is to estimate for every later node the probability $\hat{y}_v = f_{\theta}(\mathcal{G}, v) \in [0, 1]$ that it is illicit. At the same time the model returns an explanation $\mathcal{X}_v$: a ranking of the node's own features and a ranking of its incoming edges by their contribution to $\hat{y}_v$. The decision threshold on $\hat{y}_v$ is chosen on validation data (Section 4.4).

3.2. Hierarchical Recency-Aware Positional Encoding (HRAPE)

Temporal patterns are critical in fraud detection—fraudsters exploit velocity attacks (rapid transaction sequences), dormancy reactivation (sudden activity after inactivity), and coordinated timing. Unlike existing temporal GNNs

that use simple timestamp concatenation or single-scale encodings, HRAPE captures multi-scale temporal dynamics through a hierarchical encoding scheme combining recency weighting and periodic patterns. HRAPE supplies three kinds of temporal information to SEAL. It depends only on time *differences* and on counts in trailing windows, never on the absolute timestamp, so a model evaluated on future time steps never sees an out-of-range time value. In our ablation (Section 5.3), additionally encoding the absolute timestamp lowers the mean test F1 and makes it markedly less stable across seeds.

3.2.1. Multi-Scale Temporal Features

For each edge $e_{ij} = (v_i, v_j, t_{ij})$ with timestamp t_{ij} , we compute temporal features at multiple scales:—

Multi-scale gap encoding. For a set of periods $P_1, \dots, P_S$ (in the time unit of the data) the gap of an edge is encoded as

$$\phi(\Delta) = [\sin(2\pi\Delta/P_s), \cos(2\pi\Delta/P_s)]_{s=1}^S, \quad \mathbf{e}_{uv}^\tau = \mathbf{W}_\Delta \phi(\Delta_{uv}), \quad (1)$$

with $\mathbf{W}_\Delta \in \mathbb{R}^{D \times 2S}$ learnable. We use $P = (2, 4, 13, 26)$ time steps for Elliptic (one month to one year) and $P = (1, 7, 30, 90)$ days for the synthetic benchmark.

We model temporal decay using an exponential function with learnable decay rate $\lambda \in \mathbb{R}^+$:—

$$r(t_{ij}, t) = \exp(-\lambda(t - t_{ij}))$$

Recency. A learnable decay turns the gap into a recency score that biases attention (Eq. 8):

$$\rho_{uv} = \exp(-\text{softplus}(\lambda) \Delta_{uv}). \quad (2)$$

where t is the current time step. This ensures recent transactions receive higher weights, critical for detecting burst patterns.—

Financial fraud exhibits cyclical patterns (e.g., end-of-month salary transfers, weekend dormancy). We capture periodicity at multiple granularities using

sinusoidal functions:–

$$\begin{aligned}\underline{\mathbf{p}_{\text{hour}}(t)} &= \left[\sin\left(\frac{2\pi \cdot h(t)}{24}\right), \cos\left(\frac{2\pi \cdot h(t)}{24}\right) \right] \\ \underline{\mathbf{p}_{\text{day}}(t)} &= \left[\sin\left(\frac{2\pi \cdot d(t)}{7}\right), \cos\left(\frac{2\pi \cdot d(t)}{7}\right) \right] \\ \underline{\mathbf{p}_{\text{month}}(t)} &= \left[\sin\left(\frac{2\pi \cdot m(t)}{30}\right), \cos\left(\frac{2\pi \cdot m(t)}{30}\right) \right]\end{aligned}$$

Velocity. Burst behaviour is captured by the number of transactions a node received in trailing windows $w_1 < w_2 < w_3$ before τ_v ,

$$\underline{\mathbf{c}_v} = \left[\log\left(1 + |\{(u \rightarrow v) : 0 \leq \tau_v - t_{uv} < w_k\}|\right) \right]_{k=1}^3, \quad (3)$$

where $h(t)$, $d(t)$, and $m(t)$ extract hour-of-day, day-of-week, and day-of-month from timestamp t . with $w = (1, 2, 4)$ steps for Elliptic and $(1, 7, 30)$ days for the synthetic data. The velocity vector enters the input embedding (Eq. 5).

We compute transaction velocities at multiple time windows to detect burst patterns:–

$$\begin{aligned}\underline{v_{1h}(v_i, v_j)} &= \frac{\text{count}\{e \in \mathcal{E} : \text{src}(e) = v_i, \text{tgt}(e) = v_j, t - t_e \leq 1h\}}{1} \\ \underline{v_{24h}(v_i, v_j)} &= \frac{\text{count}\{e \in \mathcal{E} : \text{src}(e) = v_i, \text{tgt}(e) = v_j, t - t_e \leq 24h\}}{24} \\ \underline{v_{7d}(v_i, v_j)} &= \frac{\text{count}\{e \in \mathcal{E} : \text{src}(e) = v_i, \text{tgt}(e) = v_j, t - t_e \leq 7d\}}{168}\end{aligned}$$

What HRAPE can and cannot do on Elliptic. In Elliptic every edge connects two transactions of the same time step and each transaction appears in exactly one time step (Weber et al., 2019). Hence $\Delta_{uv} = 0$ for all edges: the gap code is a constant, $\rho_{uv} = 1$, and the velocity vector reduces to the node’s in-degree within its time step. On Elliptic HRAPE therefore contributes little beyond a degree feature, and our ablation (Section 5.3) confirms this. The temporal mechanisms become informative only on data with continuous timestamps. We test this on the synthetic benchmark (Section 5.8), where

laundering rings differ from decoy cycles only in the timing of their hops. Hour-of-day or day-of-week encodings cannot be computed from the public Elliptic data and are therefore not used.

3.3. Self-Explainable Aggregation Layer (SEAL)

Input feature mask and embedding. A two-layer MLP $g_f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ produces a per-node, per-feature mask that gates the raw input features:

$$\mathbf{M}_v^f = \sigma(g_f(\mathbf{x}_v) + 2) \in (0, 1)^d, \quad (4)$$

$$\mathbf{h}_v^0 = \text{ReLU}(\mathbf{W}_{in}(\mathbf{x}_v \odot \mathbf{M}_v^f) + \mathbf{W}_c \mathbf{c}_v). \quad (5)$$

~~We also compute acceleration (velocity change rate) to detect sudden shifts:–~~

$$a(v_i, v_j) = \frac{v_{1h}(v_i, v_j) - v_{24h}(v_i, v_j)}{v_{24h}(v_i, v_j) + \epsilon}$$

~~where $\epsilon = 10^{-6}$ prevents division by zero~~ The constant $+2$ initialises the mask close to one, so that training starts from the unmasked model. The feature mask is applied to the raw input features, where it is an attribution over identifiable inputs, and not to hidden representations.

3.3.1. Learnable Hierarchical Encoding

Masks of layer ℓ . Given $\mathbf{h}^{\ell-1}$, each SEAL layer computes a node (source) mask and an edge mask.

$$\mathbf{M}_u^n = \sigma(g_n(\mathbf{h}_u^{\ell-1})), \quad (6)$$

$$\mathbf{M}_{uv}^e = \sigma(\mathbf{w}^\top \text{ReLU}(\mathbf{A}\mathbf{h}_v^{\ell-1} + \mathbf{B}\mathbf{h}_u^{\ell-1} + \mathbf{C}_{r_{uv}} + \mathbf{T}\mathbf{e}_{uv}^\tau) + b), \quad (7)$$

where g_n is a two-layer MLP and $\mathbf{A}, \mathbf{B}, \mathbf{T} \in \mathbb{R}^{D \times D}$, $\mathbf{C}_r \in \mathbb{R}^D$. Eq. (7) is an MLP applied to the concatenation $[\mathbf{h}_v \parallel \mathbf{h}_u \parallel \mathbf{C}_r \parallel \mathbf{e}_{uv}^\tau]$, written in a factorised form that is evaluated with node-level matrix products.

~~We combine raw temporal features through learnable transformations at each scale $s \in \{\text{hour}, \text{day}, \text{week}, \text{month}\}$:–~~

$$\mathbf{T}\mathbf{E}_s(t) = \mathbf{W}_s \phi_s(t) + \mathbf{b}_s$$

Attention with recency bias. With H heads of size $D_k = D/H$,

$$\begin{aligned}\underline{\mathbf{q}}_v &= \mathbf{W}_Q \mathbf{h}_v^{\ell-1}, & \underline{\mathbf{k}}_{uv} &= \mathbf{W}_K \mathbf{h}_u^{\ell-1} + \mathbf{E}_{r_{uv}}^K + \mathbf{e}_{uv}^\tau, \\ \underline{\mathbf{m}}_{uv} &= \mathbf{W}_V \mathbf{h}_u^{\ell-1} + \mathbf{E}_{r_{uv}}^V + \mathbf{e}_{uv}^\tau, \\ \underline{\alpha}_{uv}^{(i)} &= \text{softmax}_{u \in \mathcal{N}(v)} \left(\underline{\mathbf{q}}_v^{(i)\top} \underline{\mathbf{k}}_{uv}^{(i)} / \sqrt{D_k} + \gamma \rho_{uv} \right), & i &= 1, \dots, H,\end{aligned}\quad (8)$$

where $\phi_s(t)$ denotes the Fourier features at scale s , and $\mathbf{W}_s \in \mathbb{R}^{d_{te} \times d_\phi}$, $\mathbf{b}_s \in \mathbb{R}^{d_{te}}$ are learnable parameters. The final temporal encoding aggregates across scales:—

$$\underline{\mathbf{TE}}(t) = \sum_{s \in \{\text{hour, day, week, month}\}} \mathbf{TE}_s(t)$$

⁽ⁱ⁾ selects the coordinates of head i and γ is learnable.

Gated aggregation and update. The masks gate the attention-weighted messages after normalisation, so an edge with $M_{uv}^e \rightarrow 0$ is removed from the computation instead of being compensated by a renormalised softmax:

$$\underline{\mathbf{a}}_v = \mathbf{W}_O \left\|_{i=1}^H \sum_{u \in \mathcal{N}(v)} \alpha_{uv}^{(i)} M_{uv}^e M_u^n \mathbf{m}_{uv}^{(i)}, \right. \quad (9)$$

$$\underline{\tilde{\mathbf{h}}}_v = \text{LN}(\mathbf{h}_v^{\ell-1} + \text{Dropout}(\mathbf{a}_v)), \quad \mathbf{h}_v^\ell = \text{LN}(\tilde{\mathbf{h}}_v + \text{FFN}(\tilde{\mathbf{h}}_v)). \quad (10)$$

This hierarchical design allows the model to learn which temporal scales are most relevant for fraud detection, adapting to dataset-specific patterns

Prediction and explanation. After L layers the prediction uses both the input embedding and the propagated representation (a skip connection that matters on Elliptic, where the node’s own features are highly informative):

$$\underline{\hat{y}}_v = \sigma(\text{MLP}([\mathbf{h}_v^0 \parallel \mathbf{h}_v^L])). \quad (11)$$

The explanation of v is read from the same forward pass:

$$\underline{\mathcal{X}}_v = \left(\text{TopK}_{k_f}(\mathbf{M}_v^f), \text{TopK}_{k_e} \{I_{uv} = M_{uv}^e M_u^n \bar{\alpha}_{uv} : u \in \mathcal{N}(v)\} \right), \quad (12)$$

with $\bar{\alpha}_{uv}$ the head-averaged attention of the last layer. The first part ranks the node’s own features, the second ranks its incoming (and, through the reversed copies, outgoing) transactions. Because the edge importance I_{uv} is the gate that multiplied the message of u in Eq. (9), an edge with a small I_{uv} had a proportionally small influence on $\mathbf{h}_v^L$.

3.3.1. *Recency-Weighted Attention*

3.4. *Training objective*

~~HRAPE influences message-passing through recency-weighted attention. For an edge (v_i, v_j) with timestamp t_{ij} , the attention coefficient is computed as:–~~

$$\alpha_{ij} = \frac{\exp\left(\frac{(\mathbf{W}_Q \mathbf{h}_i)^\top (\mathbf{W}_K \mathbf{h}_j)}{\sqrt{d_k}} + \gamma \cdot r(t_{ij}, t)\right)}{\sum_{v_k \in \mathcal{N}(v_i)} \exp\left(\frac{(\mathbf{W}_Q \mathbf{h}_i)^\top (\mathbf{W}_K \mathbf{h}_k)}{\sqrt{d_k}} + \gamma \cdot r(t_{ik}, t)\right)}$$

Prediction loss. We use a class-balanced binary cross-entropy in which each class contributes half of the loss.

$$\mathcal{L}_{pred} = \frac{1}{|\mathcal{V}_{tr}|} \sum_{v \in \mathcal{V}_{tr}} \left[\frac{y_v}{2\pi} \ell(\hat{y}_v, 1) + \frac{1-y_v}{2(1-\pi)} \ell(\hat{y}_v, 0) \right], \quad (13)$$

~~where $\mathbf{W}_Q, \mathbf{W}_K \in \mathbb{R}^{d_k \times d}$ are query and key projection matrices, $\mathbf{h}_i, \mathbf{h}_j$ are node embeddings, ℓ is the log-loss and $\gamma \in \mathbb{R}^+$ is a learnable temperature parameter controlling the influence of recency. This formulation ensures that recent transactions receive higher attention weights during aggregation, enabling the model to prioritize temporally relevant information.–~~

~~The core innovation of RTXGNN is the Self-Explainable Aggregation Layer (SEAL), which generates explanations intrinsically during message-passing rather than requiring post-hoc optimization. SEAL jointly learns node representations and differentiable importance masks through three parallel mask generators operating at different granularities.–~~

3.4.1. *Architecture Overview*

π the fraction of illicit labels among the training nodes $\mathcal{V}_{tr}$. Focal loss (Lin et al., 2017) is evaluated as an ablation. Its values are two orders of magnitude smaller than those of the explanation terms below, which makes the weighting of the objective fragile.

~~Each SEAL layer $\ell \in \{1, \dots, L\}$ maintains three types of learnable–~~

Explanation views. Let $\text{top}_k(\cdot)$ be the indicator of the k largest entries: the k_f largest features of $\mathbf{M}_v^f$, and for every node and layer its k_e incoming edges with the largest M_{uv}^e . From the soft masks we build two hard views, using the straight-through estimator $\mathbf{M} + (\text{top}_k(\mathbf{M}) - \mathbf{M})_{\text{no-grad}}$ so that gradients reach the mask generators:

- $g_\ell^{\text{feat}} : \mathbb{R}^{d_\ell} \rightarrow [0, 1]^{d_\ell}$ identifies which input features are important;
- $g_\ell^{\text{edge}} : \mathbb{R}^{2d_\ell} \rightarrow [0, 1]$ determines which edges contribute to predictions the *explanation view S* keeps only the selected features and edges (all other gates are set to zero);
- $g_\ell^{\text{node}} : \mathbb{R}^{d_\ell} \rightarrow [0, 1]$ scores the importance of each node in the computation graph the *complement view C* keeps only the non-selected features and edges.

Each generator is implemented as a two-layer MLP with sigmoid activation:

$$g(\mathbf{x}) = \sigma(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2)$$

where σ is the sigmoid function ensuring outputs lie in $[0, 1]$. Running the model on the two views gives $\hat{y}_w^S$ and $\hat{y}_w^C$ (M^n is kept soft in both views).

3.4.1. Forward Pass with Mask Generation

The forward pass of SEAL layer ℓ proceeds as follows:

Given input embeddings $\mathbf{H}^{(\ell-1)} \in \mathbb{R}^{n \times d_{\ell-1}}$, generate feature masks:

$$\mathbf{M}^{\text{feat},(\ell)} = g_\ell^{\text{feat}}(\mathbf{H}^{(\ell-1)})$$

Apply masks element-wise to identify important features:

$$\tilde{\mathbf{H}}^{(\ell-1)} = \mathbf{H}^{(\ell-1)} \odot \mathbf{M}^{\text{feat},(\ell)}$$

where $\odot$ denotes element-wise multiplication.

For each edge $(v_i, v_j) \in \mathcal{E}$, compute edge representation:

$$\mathbf{e}_{ij} = [\tilde{\mathbf{h}}_i^{(\ell-1)} \parallel \tilde{\mathbf{h}}_j^{(\ell-1)}]$$

where $\parallel$ denotes concatenation.

Generate edge mask and temporal weight:-

$$\underline{m_{ij}^{\text{edge}} = g_{\ell}^{\text{edge}}(\mathbf{e}_{ij})}$$

$$\underline{w_{ij}^{\text{temp}} = \text{TemporalScorer}(\mathbf{TE}(t_{ij}))}$$

The temporal scorer is a learned function mapping temporal encodings to importance weights:-

$$\underline{\text{TemporalScorer}(\mathbf{TE}) = \sigma(\mathbf{w}_{\text{temp}}^{\top} \mathbf{TE} + b_{\text{temp}})}$$

Compute masked message with temporal weighting:-

$$\underline{\mathbf{m}_{ji \rightarrow i} = \mathbf{W}^{(\ell)} \tilde{\mathbf{h}}_j^{(\ell-1)} \cdot m_{ij}^{\text{edge}} \cdot w_{ij}^{\text{temp}}}$$

Sufficiency and necessity. With $\text{KL}(p||q)$ the Kullback-Leibler divergence between Bernoulli distributions,

$$\underline{\mathcal{L}_{\text{suf}} = \frac{1}{|\mathcal{V}_{tr}|} \sum_{v \in \mathcal{V}_{tr}} \text{KL}(\text{sg}(\hat{y}_v) \parallel \hat{y}_v^S)}, \quad (14)$$

$$\underline{\mathcal{L}_{\text{nec}} = \frac{1}{|\mathcal{V}_{tr}|} \sum_{v \in \mathcal{V}_{tr}} \text{KL}(\pi \parallel \hat{y}_v^C)}, \quad (15)$$

where $\mathbf{W}^{(\ell)} \in \mathbb{R}^{d_{\ell} \times d_{\ell-1}}$ is a learnable transformation matrix.-

Aggregate messages using multi-head attention with recency-weighted coefficients from HRAPE:-

$$\underline{\mathbf{h}_i^{(\ell)} = \text{MultiHeadAttn} \left(\tilde{\mathbf{h}}_i^{(\ell-1)}, \{\mathbf{m}_{ji \rightarrow i}\}_{v_j \in \mathcal{N}(v_i)}, \{\alpha_{ij}\}_{v_j \in \mathcal{N}(v_i)} \right)}$$

Generate node importance scores from updated embeddings:-

$$\underline{\mathbf{M}^{\text{node},(\ell)} = g_{\ell}^{\text{node}}(\mathbf{H}^{(\ell)})}$$

The final layer embeddings are :-

$$\underline{\tilde{\mathbf{H}}^{(\ell)} = \mathbf{H}^{(\ell)} \odot \mathbf{M}^{\text{node},(\ell)}}$$

3.4.1. ~~Explanation Aggregation Across Layers~~

~~After L layers, we aggregate masks across depths to obtain final explanations:~~

$$\begin{aligned}\underline{\mathbf{M}_{\text{final}}^{\text{feat}}} &= \frac{1}{L} \sum_{\ell=1}^L \mathbf{M}^{\text{feat},(\ell)} \\ \underline{\mathbf{M}_{\text{final}}^{\text{edge}}} &= \frac{1}{L} \sum_{\ell=1}^L \mathbf{M}^{\text{edge},(\ell)} \\ \underline{\mathbf{M}_{\text{final}}^{\text{node}}} &= \frac{1}{L} \sum_{\ell=1}^L \mathbf{M}^{\text{node},(\ell)}\end{aligned}$$

~~This aggregation ensures that explanations capture information flow across all network depths, not just the final layers~~ g stops the gradient. $\mathcal{L}_{\text{suf}}$ asks that the top- k explanation alone reproduce the model’s prediction. $\mathcal{L}_{\text{nec}}$ asks that the remaining inputs alone carry no information about the label, i.e. that the prediction on the complement fall back to the class prior. These are the factual and counterfactual criteria of Tan et al. (2022), and they match the Fid^- and Fid^+ metrics used in evaluation (Yuan et al., 2021; Amara et al., 2022)

~~The final layer embeddings $\tilde{\mathbf{H}}^{(L)}$ are processed by two parallel heads:–
Maps node embeddings to fraud probabilities:–~~

$$\underline{\hat{y}_i = \sigma(\mathbf{w}_{\text{pred}}^\top \tilde{\mathbf{h}}_i^{(L)} + b_{\text{pred}})}$$

~~Generates human-interpretable explanations by identifying–~~

Why the objective cannot be met by ignoring the masks. A fidelity loss of the form $\text{KL}(f(\mathcal{G}) \| f(\mathcal{G}_{\text{masked}}))$ with soft masks alone is minimised trivially by masks that are all close to one, or by a model that is insensitive to the masks, and says nothing about whether the masks are faithful. Our objective rules out both degenerate solutions: (i) the views are *hard*: in S all but k_f of the d features and all but k_e incoming edges are removed, so $\mathcal{L}_{\text{suf}}$ can only be small if the selected elements actually carry the prediction; (ii) a model that ignores the masks predicts the same value on the complement C as on the full input, which incurs $\mathcal{L}_{\text{nec}} > 0$ for every node whose prediction differs from

the prior; (iii) a mask that is uninformative about importance (for instance uniform) produces a random top- K important elements:–

$$\begin{aligned}\mathcal{F}_{\text{top-K}} &= \text{TopK}(\mathbf{M}_{\text{final}}^{\text{feat}}, K_{\text{feat}}) \\ \mathcal{E}_{\text{top-K}} &= \text{TopK}(\mathbf{M}_{\text{final}}^{\text{edge}}, K_{\text{edge}}) \\ \mathcal{V}_{\text{top-K}} &= \text{TopK}(\mathbf{M}_{\text{final}}^{\text{node}}, K_{\text{node}})\end{aligned}$$

These sparse explanations directly indicate which features, edges, and neighbors drove the prediction, enabling regulatory-compliant interpretability.

RTXGNN is trained end-to-end using a multi-task objective that jointly optimizes prediction accuracy, explanation fidelity, and explanation sparsity.

3.4.1. Loss Functions

We use focal loss (Lin et al., 2017) to handle severe class imbalance in fraud detection:–

$$\mathcal{L}_{\text{pred}} = -\frac{1}{n} \sum_{i=1}^n \alpha_i (1 - \hat{y}_i)^\gamma y_i \log(\hat{y}_i) + \alpha_i \hat{y}_i^\gamma (1 - y_i) \log(1 - \hat{y}_i)$$

where α_i is the class weight for node v_i , and $\gamma = 2$ is the focusing parameter that down-weights easy examples.–

Ensures that masked predictions match full model predictions, guaranteeing that explanations faithfully represent k selection, for which both terms are large unless the model does not depend on the inputs at all, and that case is excluded by $\mathcal{L}_{\text{pred}}$. The objective uses only the model’s reasoning:–

$$\mathcal{L}_{\text{fidelity}} = \frac{1}{n} \sum_{i=1}^n \text{KL} (f_\theta(\mathcal{G}, v_i) \| f_\theta(\mathcal{G}_{\text{masked}}^i, v_i))$$

where $\mathcal{G}_{\text{masked}}^i$ retains only top- K masked elements for node v_i , and KL is Kullback-Leibler divergence own predictions and the class prior, never the labels of the explained nodes, so it does not supervise the explanation with labels.

~~Encourages concise explanations by penalizing high mask values:–~~

$$\mathcal{L}_{\text{sparse}} = \frac{1}{n \cdot d_v} \sum_{i=1}^n \|\mathbf{M}_i^{\text{feat}}\|_1 + \frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} m_{ij}^{\text{edge}} + \frac{1}{n} \sum_{i=1}^n m_i^{\text{node}}$$

~~The ℓ_1 penalty promotes sparse masks, ensuring explanations highlight only the most critical elements.–~~

~~Ensures temporal importance weights align with fraud timing patterns by maximizing separation between fraud and benign transaction temporal scores:–~~

$$\mathcal{L}_{\text{temp}} = -\frac{1}{|\mathcal{E}_{\text{fraud}}|} \sum_{e \in \mathcal{E}_{\text{fraud}}} w_e^{\text{temp}} + \frac{1}{|\mathcal{E}_{\text{benign}}|} \sum_{e \in \mathcal{E}_{\text{benign}}} w_e^{\text{temp}}$$

Sparsity. $\mathcal{L}_{sp}$ is the mean L_1 norm of all masks,

$$\mathcal{L}_{sp} = \overline{\mathbf{M}^f} + \frac{1}{L} \sum_{\ell=1}^L (\overline{M^{e,(\ell)}} + \overline{M^{n,(\ell)}}), \quad (16)$$

where the bar denotes the mean over all entries. This is ordinary L_1 regularisation of auxiliary gate activations. We do not claim it is a new form of regularisation, and Section 5.4 compares it with dropout and weight decay.

~~This loss encourages the model to assign higher temporal importance to transactions during periods associated with fraud.–~~

3.4.1. *Combined Objective with Curriculum Learning*

~~The total training objective combines all loss terms:–~~

$$\mathcal{L}_{\text{total}} = \lambda_{\text{pred}} \mathcal{L}_{\text{pred}} + \lambda_{\text{fid}} \mathcal{L}_{\text{fidelity}} + \lambda_{\text{sparse}} \mathcal{L}_{\text{sparse}} + \lambda_{\text{temp}} \mathcal{L}_{\text{temp}}$$

No label supervision of the gates. We deliberately do not supervise any mask or temporal gate with labels (for example by rewarding high gates on edges incident to fraudulent nodes). Such a term would leak label information into what is presented as an explanation.

~~To stabilize training–~~

Total objective and curriculum.

$$\mathcal{L} = \mathcal{L}_{pred} + w(e)(\lambda_{suf}\mathcal{L}_{suf} + \lambda_{nec}\mathcal{L}_{nec} + \lambda_{sp}\mathcal{L}_{sp}), \quad w(e) = \min(1, e/e_w), \quad (17)$$

with epoch e , we employ curriculum learning by gradually increasing explanation-related loss weights:—

$$\lambda_{fid}(e) = \lambda_{fid}^{\max} \cdot \min\left(1, \frac{e}{e_{warmup}}\right), \quad \lambda_{sparse}(e) = \lambda_{sparse}^{\max} \cdot \min\left(1, \frac{e}{e_{warmup}}\right)$$

where e is the current epoch and e_{warmup} is the warmup period (typically 10-20 epochs) warm-up $e_w = 20$, $\lambda_{suf} = \lambda_{nec} = 0.5$, $\lambda_{sp} = 0.05$, $k_f = 10$ and $k_e = 2$. These values were fixed before the experiments reported here and were not tuned on test data. This allows the model to first learn accurate predictions before enforcing explainability constraints.

3.4.1. Training Algorithm

3.5. Training and inference procedures

Algorithm 1 summarizes the complete training procedure Algorithm 1 gives the training procedure. We train full-batch on the subgraph induced by the 2-hop neighbourhoods of the labelled training nodes. For a model with $L = 2$ message-passing layers this is exact: every training prediction depends only on the 2-hop neighbourhood of its node, so no neighbourhood sampling is used during training. Evaluation on validation and test nodes uses the 2-hop neighbourhoods of those nodes in the same way.

At inference time (Algorithm 2) a request consists of one or more transactions to score. For each target we sample at most $F = 25$ incoming neighbours per hop for two hops from an in-memory CSR adjacency, run one forward pass on the union of the sampled subgraphs, and read the explanation of every target from the masks of that pass (Eq. 12). On Elliptic the in-degree rarely exceeds F , so the sampled neighbourhood is almost always the exact 2-hop neighbourhood. The latency of every step of this procedure is measured in Section 5.10.

3.5.1. Computational Complexity

The computational complexity of RTXGNN per layer is $\mathcal{O}(|\mathcal{E}| \cdot d_t \cdot d_{t+1})$ for message computation and aggregation, plus $\mathcal{O}(n \cdot d_t^2)$ for mask generation

Algorithm 1 RTXGNN training (as implemented in `rtxgnn/train.py` and `rtxgnn/objective.py`).

Require: graph $\mathcal{G}$ with features $\mathbf{X}$, timestamps; labelled training nodes $\mathcal{V}_{tr}$ with labels y ; validation nodes $\mathcal{V}_{val}$; hyper-parameters $\lambda_{suf}, \lambda_{nec}, \lambda_{sp}, k_f, k_e, e_w, E$, patience Q

Ensure: parameters θ^* and decision threshold η^*

- 1: $\mathcal{G}_{tr} \leftarrow$ subgraph induced by the 2-hop neighbourhoods of $\mathcal{V}_{tr}$; $\mathcal{G}_{ev} \leftarrow$ same for $\mathcal{V}_{val}$
 - 2: compute Δ_{uv} for all directed edges and $\mathbf{c}_v$ for all nodes (Eqs. 1-3); $\pi \leftarrow$ mean of y over $\mathcal{V}_{tr}$
 - 3: initialise θ ; Adam optimiser (learning rate $5 \cdot 10^{-3}$, weight decay 10^{-5})
 - 4: **for** $e = 0, \dots, E - 1$ **do**
 - 5: $\mathbf{M}^f, \{M^{n,(\ell)}, M^{e,(\ell)}\}_{\ell}, \hat{\mathbf{y}} \leftarrow$ forward pass of Eqs. 4-11 on $\mathcal{G}_{tr}$
 - 6: $\mathcal{L} \leftarrow \mathcal{L}_{pred}(\hat{\mathbf{y}}_{\mathcal{V}_{tr}}, y)$ {Eq. 13}
 - 7: $w \leftarrow \min(1, e/e_w)$
 - 8: **if** $w > 0$ **then**
 - 9: $S \leftarrow$ hard top- k_f features per node and top- k_e incoming edges per node and layer (straight-through); $C \leftarrow 1 - S$
 - 10: $\hat{\mathbf{y}}^S \leftarrow$ forward pass with masks replaced by S ; $\hat{\mathbf{y}}^C \leftarrow$ forward pass with masks replaced by C
 - 11: $\mathcal{L} \leftarrow \mathcal{L} + w \lambda_{suf} \mathcal{L}_{suf} + w \lambda_{nec} \mathcal{L}_{nec}$ {Eqs. 14, 15}
 - 12: **end if**
 - 13: $\mathcal{L} \leftarrow \mathcal{L} + w \lambda_{sp} \mathcal{L}_{sp}$ {Eq. 16}
 - 14: $\theta \leftarrow \theta - \text{Adam}(\nabla_{\theta} \mathcal{L})$ with gradient-norm clipping at 1
 - 15: **if** e is even **then**
 - 16: compute average precision AP_{val} of the forward pass on $\mathcal{G}_{ev}$ at the nodes $\mathcal{V}_{val}$
 - 17: **if** AP_{val} improved **then** $\theta^* \leftarrow \theta$ **else if** no improvement for Q epochs **then break**
 - 18: **end if**
 - 19: **end for**
 - 20: $\eta^* \leftarrow \arg \max_{\eta} \text{F1}_{val}(\eta)$ for the model θ^*
 - 21: **return** θ^*, η^*
-

– The temporal encoding adds $\mathcal{O}(|\mathcal{E}| \cdot d_{te})$. For typical settings ($L = 3$ layers, $d_t = 128$, $d_{te} = 64$), this remains tractable for real-time inference on modern GPUs, with per-transaction latency under 50ms as demonstrated in Section 5

Algorithm 2 RTXGNN inference for a request (as implemented in `rtxgnn/serving.py`).

Require: trained θ^* , threshold η^* ; CSR adjacency of the live graph; target transactions B ; fan-out $F = 25$; k_f, k_e

- 1: $N_0 \leftarrow B$; $\mathcal{E}_B \leftarrow \emptyset$
 - 2: **for** hop $h = 1, 2$ **do**
 - 3: for every $v \in N_{h-1}$ draw at most F incoming edges uniformly at random; add them to $\mathcal{E}_B$
 - 4: $N_h \leftarrow$ sources of the edges drawn in this hop
 - 5: **end for**
 - 6: build the subgraph on $N_0 \cup N_1 \cup N_2$ with edges $\mathcal{E}_B$ (features, Δ_{uv} , types, $\mathbf{c}_v$)
 - 7: one forward pass (Eqs. 4–11) without gradients
 - 8: **for** $v \in B$ **do**
 - 9: score $\hat{y}_v$; decision $\mathbb{K}[\hat{y}_v \geq \eta^*]$; explanation $\mathcal{X}_v$ from Eq. 12
 - 10: **end for**
 - 11: **return** scores, decisions and explanations
-

Complexity. A SEAL layer costs $\mathcal{O}(|\mathcal{V}|D^2 + |\mathcal{E}|D)$: the projections and mask MLPs are node-level matrix products, and each edge requires $\mathcal{O}(D)$ operations for its attention score, gate and message. Feature-mask generation adds $\mathcal{O}(|\mathcal{V}|dD)$. The explanation adds only a top- k selection at inference time. During training each epoch performs three forward passes (full, S and C), so an epoch costs about three times as much as for a SEAL network without the explanation objective.

~~We implement RTXGNN in PyTorch~~

Implementation. RTXGNN is implemented in PyTorch (Paszke et al., 2019) with PyTorch Geometric ~~–(Fey et al., 2025) for efficient graph operations.~~ ~~Key hyperparameters are: $L = 3$ GNN (Fey et al., 2025). We use $L = 2$ SEAL layers, hidden dimension $d = 128$, 4 attention size $D = 64$, $H = 4$ heads, dropout rate 0.3, batch size 256, learning rate 0.001 with cosine annealing, and $\lambda_{\text{pred}} = 1.0$, $\lambda_{\text{fid}}^{\text{max}} = 0.1$, $\lambda_{\text{sparse}}^{\text{max}} = 0.05$, $\lambda_{\text{temp}} = 0.01$. Models are trained for 200 epochs with early stopping (patience = 20) on validation F1-score. We use 4-hop neighborhood sampling with maximum 50 neighbors per hop for scalability. All experiments use NVIDIA A100 GPUs with mixed-precision training (FP16) to accelerate convergence. 0.2, at most 300~~

epochs, early stopping with patience $Q = 40$ epochs on validation average precision (evaluated every second epoch), and gradient clipping at norm 1. Hardware is described in Section 4.4.

RTXGNN differs from prior temporal GNN frameworks in that it treats temporal encoding and intrinsic explainability as jointly designed components rather than independent additions. Existing approaches such as Time2Vec (?) and TGAT (?) embed temporal information at a single scale without connecting it to any explanation mechanism, while self-explainable models such as SEFraud (Li et al., 2024) generate interpretable masks on static graphs with no temporal awareness. RTXGNN unifies these two directions: HRAPE provides multi-scale temporal signals—recency, periodicity, and velocity—that are directly consumed by SEAL’s mask generators, so that the resulting explanations reflect not only which features and neighbors matter, but *when* they matter.

4. Experimental setup

This section presents a comprehensive evaluation of RTXGNN across multiple datasets and experimental conditions. We compare RTXGNN against state-of-the-art baselines including temporal GNNs and fraud-specific methods (Section ??), conduct extensive ablation studies (Section 5.3), analyze label efficiency and temporal stability (Sections ?? and ??), quantify explanation quality (Section ??), and present interpretable case studies (Section ??). We evaluate on three datasets: Elliptic Bitcoin (primary benchmark), Amazon Computers (generalization test), and Synthetic Money Laundering Network (controlled evaluation). All numbers in this and the following sections are produced by the scripts in the `experiments/` directory of the public repository. Each script writes one JSON record per run, and the tables are generated from these records.

4.0.1. Datasets

4.1. Datasets

We evaluate on the Elliptic Bitcoin dataset (Weber et al., 2019), which *Elliptic Bitcoin dataset*. The Elliptic dataset (Weber et al., 2019) contains 203,769 Bitcoin transactions (nodes) and 234,355 directed payment flows (edges) spanning in 49 time steps. Each transaction is labeled as *licit* (42%), *illicit* (2%) of about two weeks each, or *unknown* (56%). The

transactions (2.2%) are labelled illicit, 42,019 (20.6%) licit and the remaining 157,205 (77.1%) are unlabelled. Each transaction has 166 node features include local information (transaction timestamp, input/output counts, fees) and aggregate features (maximum/minimum/average/standard deviation of neighbor transaction values across 1-hop and 2-hop neighborhoods) anonymised features. The first is the time step. The next 93 are *local* features of the transaction (the dataset documentation mentions, among others, the numbers of inputs and outputs, fees and output volume). The dataset exhibits severe class imbalance (licit:illicit ratio $\approx 21:1$) and temporal evolution, making it ideal for evaluating fraud detection systems under realistic conditions.

Following the temporal evaluation protocol recommended by (Weber et al., 2019), we use time-based splits: $\therefore$ Time last 72 are *aggregated* features, statistics over the transaction’s one-hop neighbourhood. The time step is not used as a feature, so every model receives $d = 165$ features. No edge crosses a time step. We use a temporal split: steps 1–30 (157,205 transactions) $\therefore$ Time for training (26,905 labelled transactions, 2,954 illicit), steps 31–34 (18,261 transactions) $\therefore$ Time for validation (2,989; 508 illicit) and steps 35–49 (28,303 transactions) This split ensures evaluation on temporally distant future transactions, assessing ability to handle concept drift and evolving fraud tactics for testing (16,670; 1,083 illicit). Appendix A (Table A.17) lists the number of licit, illicit and unlabelled transactions in every time step.

A co-purchase graph from the Amazon e-commerce platform, consisting of 13,752 nodes (products) and 245,861 edges. The task is to classify products into one of

T-Finance. T-Finance (Tang et al., 2022) is a transaction network of a financial company with 39,357 accounts, 21.2 million undirected edges (transactions between accounts) and 10 categories. This dataset tests the model’s ability to generalize to non-financial domains and handle multi-class classification problems.

To rigorously test the model’s ability to detect specific structural fraud patterns, we generated a synthetic graph with 5,000 nodes. This dataset injects specific "fraud ring" typologies (cyclic transaction patterns) into a scale-free base network, allowing us to verify if the model can identify and explain these known structures.

4.1.1. Baseline Methods

~~We compare RTXGNN against nine baseline methods spanning general GNNs, temporal GNNs, fraud-specific GNNs, and explainable GNNs:–~~

- ~~• (Kipf and Welling, 2017): Graph Convolutional Network with spectral graph convolutions–~~
- ~~• (Veličković et al., 2018): Graph Attention Network with learnable attention weights–~~
- ~~• (Hamilton et al., 2017): Sampling-based GNN with mean aggregation–~~
- ~~• : Multi-layer perceptron using only node features (no graph structure)–~~

~~(Pareja et al., 2020): Evolves GNN parameters over time using RNN account features; 4.58% of the accounts are labelled fraudulent by human experts. We use the version and the first five fully supervised splits (40%/LSTM, capturing temporal dynamics through weight evolution (Sankar et al., 2020) : Dynamic self-attention on both structural and temporal dimensions–~~

- ~~• (Dou et al., 2020): Addresses camouflaged fraudsters through reinforcement learning-based neighbor selection–~~
- ~~• (Liu et al., 2021): Tackles class imbalance through pick-and-choose sampling strategy–~~

20%/40% train/validation/test) of GADBench (Tang et al., 2023). T-Finance has no timestamps, so it tests whether SEAL and the explanation objective carry over to a second real-world financial network with different features, entities and label source. HRAPE is switched off and temporal baselines are not applicable. To keep full-batch training tractable in memory, graph models see at most 10 uniformly sampled neighbours per account, which reduces the graph to about 0.4 million undirected edges.

- ~~• (Li et al., 2024): Self-explainable fraud detection with learnable feature and edge masks (static graph only)–~~

Synthetic temporal laundering benchmark. The synthetic benchmark contains a signal that is known by construction and that only a model using timing information can exploit. The generator (`rtxgnn/synthetic.py`) works as follows. (1) A Barabási–Albert graph on $n = 5,000$ accounts ($m = 2$) defines which pairs transact. Each pair gets a random direction and a Poisson number of transactions (mean 3), with timestamps uniform on $[0, 180]$ days and log-normal amounts ($\mu = 5, \sigma = 1$). (2) Fifty laundering rings are laid on disjoint accounts. Each is a directed cycle of length $L \sim \mathcal{U}\{4, \dots, 8\}$ executed within a *burst* of 2 days at a random start time, with a 1–3% fee per hop. Ring members keep their ordinary transactions. (3) Fifty *decoy* cycles with the same length and amount distribution are laid on other accounts, but their hops are spread over the whole horizon. The cycle structure alone therefore does not identify laundering; the timing of consecutive hops does. (4) Every account has 16 features: 7 behavioural statistics computed from all its transactions (in/out counts, mean and standard deviation of in/out log-amounts, number of counterparties; standardised), a constant, and 8 i.i.d. $\mathcal{N}(0, 1)$ noise features. Ring members form the positive class (6.1% of accounts), and the ground-truth explanation of a ring member consists of its two ring transactions. All accounts are scored at $t = 180$ days with a stratified random 60/20/20 split. Besides this *default* setting we generate four variants: *no decoy* (no decoy cycles, so structure alone suffices), *wide burst* (rings completed within 10 instead of 2 days), *label noise* (5% of labels flipped) and *shift* (+0.5 added to the noise features of ring members, a weak feature signal). Every setting is generated with seeds 0–4, and seed s is also the model seed. By default the features carry no class signal, so the task cannot be solved from the features alone.

- (~~Dou et al., 2020~~): ~~Addresses camouflaged fraudsters through reinforcement learning-based neighbor selection~~
- (~~Liu et al., 2021~~): ~~Tackles class imbalance through pick-and-choose sampling strategy~~
- (~~Shi et al., 2022~~): ~~Differentiates homophilic and heterophilic edges in fraud graphs through dedicated aggregation strategies for each connection type~~
- (~~Duan et al., 2024~~): ~~Addresses non-additivity of node attributes through dynamic grouping aggregation with decision tree-based feature binning~~

4.2. Preprocessing

~~All baselines use the same architecture depth ($L = 3$ layers), hidden dimensions ($d = 128$), and training hyperparameters for fair comparison.~~ For all neural models, each feature is mapped to a standard normal distribution by a quantile transform fitted on the transactions of the training steps only (Elliptic: steps 1–30, labelled and unlabelled; T-Finance: training accounts). No labels are used in this fit. Elliptic features are heavy-tailed, and the transform improves every neural model substantially (Table C.18 in Appendix C reports results with the raw features). The transform was adopted during development, when test results were also computed. To rule out a choice that fits only the test period, we checked it on the validation steps, which are used for model selection: the transform raises the validation AP of 5 of the 6 neural models and leaves that of the sixth (SEFraud) practically unchanged. Tree ensembles and logistic regression use the raw features. The transform does not help trees, which are invariant to monotone transformations, and on Elliptic its quantisation slightly reduced the random forest’s score.

4.2.1. ~~Ablated Variants of RTXGNN~~

4.3. Baselines

~~To isolate contributions of individual components, we evaluate six ablated variants:~~ We compare against 16 baselines in four groups. Appendix B describes how each one is adapted.

- ~~• Removes Hierarchical Recency-Aware Positional Encoding, using only static node features~~
 - ~~• Removes sparsity regularization ($\lambda_{\text{sparse}} = 0$)~~
 - ~~• Replaces SEAL with standard GAT attention (no intrinsic masks)~~
- Feature-only models:** logistic regression (LR, class-balanced), random forest (RF; 100 trees, 50 features per split, the configuration of Weber et al., 2019) and XGBoost (Chen and Guestrin, 2016) (300 trees, depth 6, class-balanced), and a two-layer MLP.
- ~~• Removes temporal consistency loss ($\lambda_{\text{temp}} = 0$)~~ **General GNNs:** GCN (Kipf and Welling, 2017), GraphSAGE (Hamilton et al., 2017) and GAT (Veličković et al., 2018).

- ~~÷ Removes explanation fidelity loss ($\lambda_{\text{fd}} = 0$)~~ **Temporal GNNs:** EvolveGCN-O (Pareja et al., 2020), TGAT (Xu et al., 2020), TGN (Rossi et al., 2020) and APAN (Wang et al., 2021).
- ~~÷ Removes curriculum learning (all losses weighted equally from epoch 1)~~ **Fraud-specific GNNs:** CARE-GNN (Dou et al., 2020), PC-GNN (Liu et al., 2021), GAS (Li et al., 2019), FRAUDRE (Zhang et al., 2021), and the self-explaining SEFraud (Li et al., 2024).

All graph baselines were re-implemented in the same code base, because the original implementations target multi-relation review graphs (CARE-GNN, PC-GNN, FRAUDRE), heterogeneous graphs (GAS, SEFraud) or link prediction on continuous-time graphs (TGAT, TGN, APAN). The re-implementations follow the published architectures, adapted to the single-relation Elliptic graph. We cannot guarantee that they reach the performance of the authors' own code in the original settings.

4.3.1. ~~Evaluation Metrics~~

~~We report the following metrics:–~~

- ~~÷ Harmonic mean of precision and recall, critical for imbalanced datasets~~
- ~~÷ Measures ranking quality across all classification thresholds~~
- ~~÷ Fraction of predicted frauds that are actual frauds~~
- ~~÷ Fraction of actual frauds correctly identified~~
- ~~÷ Average mask value across all explanations (lower indicates more selective/sparse explanations)~~

DySAT is not included because its temporal attention is vacuous on Elliptic (each node appears in a single snapshot; Section 2). H2-FDetector and DGA-GNN, which were compared in the first version of this study under a different protocol, are not included either. H2-FDetector separates homophilic from heterophilic connections within each relation of a multi-relation graph, and DGA-GNN groups non-additive raw attributes by decision-tree binning. On Elliptic's single-relation graph with engineered features, their central mechanisms are represented by the label-aware neighbour selection of CARE-GNN and PC-GNN, the node-neighbour differences of FRAUDRE, and the tree ensembles, which are all included.

4.3.1. *Implementation Details*

4.4. *Protocol and statistical analysis*

All models are implemented in PyTorch 2.0 with PyTorch Geometric 2.3. Training uses Adam optimizer with initial learning rate $\eta = 0.001$, cosine annealing schedule, batch size 256, and

Training. All neural models use the same protocol and budget: hidden size 64, two message-passing layers where applicable, dropout 0.2, the class-balanced loss of Eq. (13) (plus the method’s own auxiliary losses), Adam with learning rate $5 \cdot 10^{-3}$ and weight decay 10^{-5} , gradient clipping at norm 1.0. We apply focal loss with $\gamma = 2$ and class weights computed from training set distribution to address class imbalance. Models are trained for maximum 200 epochs with early stopping (patience = 20) based on validation F1 score. For RTXGNN, we use curriculum learning with warmup period $e_{\text{warmup}} = 10$ epochs and loss weights $\lambda_{\text{pred}} = 1.0$, $\lambda_{\text{fid}}^{\text{max}} = 0.1$, $\lambda_{\text{sparse}}^{\text{max}} = 0.05$, $\lambda_{\text{temp}} = 0.01$. All experiments run on NVIDIA A100 GPUs with mixed-precision training (FP16)¹, at most 300 full-batch epochs, and early stopping on validation average precision with a patience of 40 epochs. No model, including RTXGNN, was tuned on the test data, and no baseline received a per-model hyper-parameter search. This is a limitation that affects all methods equally.

Decision threshold. Every model outputs a score. The decision threshold is the one that maximises F1 on the validation set, and it is then applied unchanged to the test set. We also report threshold-free metrics: the area under the ROC curve (AUC) and average precision (AP, the area under the precision-recall curve). AP is the more informative of the two under class imbalance.

Metrics. We report F1, precision and recall of the illicit class, together with AUC and AP. The overall test F1 is computed *once over all 16,670 labelled test transactions pooled together* (micro-averaging over time steps), as in Weber et al. (2019). It is not the mean of the per-step F1 scores. Section 5.5 explains why the two differ.

Seeds and tests. Each configuration on Elliptic is run with 10 random seeds (0–9), which change the parameter initialisation, dropout masks and, for the random forest and XGBoost, the bootstrap and column sampling. We report mean ~~results over~~ $\pm$ standard deviation over the seeds. Logistic regression is deterministic. We compare RTXGNN with every baseline with a two-sided

Wilcoxon signed-rank test on the per-seed F1 and AP pairs (Demšar, 2006). Seeds are paired because both models use the same seed. We apply Holm’s correction (Holm, 1979) over the 16 comparisons of each metric. Ablations use 5 seeds (3 runs with different random seeds).

Table ?? presents comprehensive comparison of RTXGNN against all baseline methods on the Elliptic test set. Figure ?? shows the learning dynamics across training epochs.

for the secondary variants), T-Finance uses GADBench splits 0–4, and the synthetic benchmark uses 5 generated graphs per setting. With 5 or 3 pairs, a two-sided Wilcoxon test cannot reach $p < 0.05$, so the ablations report paired t -tests. These p -values are not corrected for multiple comparisons and should be read as indicative.

The full RTXGNN model achieves F1-score of 0.5129, outperforming all baselines: +4.2% over SEFraud (best explainable baseline), +5.6% over PC-GNN (best fraud-specific baseline), +11.2% over DySAT (best temporal baseline)

Hardware. Experiments were distributed over two kinds of machines: a cloud virtual machine with 4 vCPUs of an Intel Xeon processor (Intel Xeon Processor @ 2.80GHz) and 15 GB of memory (CPU only), and +19.3% over GAT (best general GNN) Google Colab virtual machines with an NVIDIA T4 GPU. In the main comparison, seeds 0–2 and 5–9 ran on the CPU machine (except RTXGNN with seed 2) and seeds 3–4 on the GPU. The primary ablation study, the synthetic benchmark and the explanation study ran on the CPU machine. The EvolveGCN runs, the label-efficiency, retraining and T-Finance experiments, and the secondary ablations ran on the GPU. Each secondary ablation is compared with a reference model trained on the same hardware. Because the studies use different seed sets and hardware, the F1 of the full RTXGNN differs between them: 0.700 in the main comparison (10 seeds), 0.696 in the primary ablation (5 seeds, CPU), 0.724 in the secondary ablation (3 seeds, GPU) and 0.718 with all labels in the label-efficiency study (5 seeds, GPU). This demonstrates that jointly learning predictions and explanations through SEAL with temporal awareness through HRAPE does not sacrifice—and actually improves—detection accuracy compared to methods addressing individual aspects. The superior F1-score indicates RTXGNN achieves better precision-recall trade-off, critical for production deployment where both false positives (alert fatigue) and false negatives (missed fraud) carry significant costs. Each study is compared only with its own reference. Both machines used PyTorch 2 and

PyTorch Geometric 2 without mixed precision, and the algorithm is identical on both. All latency measurements (Section 5.10) were taken on the CPU machine.

RTXGNN’s AUC of 0.8661 is competitive with baselines, slightly lower than GraphSAGE (0.8735) but comparable to other strong methods. The strong AUC indicates excellent ranking quality—the model reliably assigns higher scores to fraudulent transactions across all decision thresholds. The slight AUC gap versus GraphSAGE is acceptable given RTXGNN’s superior F1-score (+23.7%) and unique combination of temporal awareness and intrinsic explainability that GraphSAGE lacks.

5. Results

EvolveGCN (

5.1. Detection performance on Elliptic

Table 3 reports the test results of all models over 10 seeds.

RTXGNN reaches a test F1 of 0.700 ± 0.045 and an AP of 0.771. Among the twelve graph neural networks, only GAS obtains a practically identical F1 (=0.4423) and DySAT (0.701), and FRAUDRE and GAS obtain a practically identical AP (0.775 and 0.773). RTXGNN is ahead of all other GNNs on F1 (=0.4612) capture temporal dynamics but lack fraud-specific design considerations (camouflage handling, class imbalance strategies). This validates that domain-specific architectures matter—RTXGNN’s superior performance stems from combining temporal modeling (HRAPE) with, including the general GNNs (GraphSAGE 0.684, GCN 0.690, GAT 0.646), the temporal GNNs (TGN 0.685, TGAT 0.673) and the fraud-specific mechanisms (sparsity regularization, focal loss).

SEFraud (models. In particular, RTXGNN improves on the other self-explaining model, SEFraud (+0.054 F1, $p = 0.010$ uncorrected; +0.029 AP, $p = 0.006$). After Holm correction over the 16 comparisons, the F1 =0.4923) achieves strong performance through intrinsic explainability but operates on static graphs. RTXGNN’s +4.2% improvement demonstrates that HRAPE’s temporal encoding captures fraud-relevant patterns that static features miss, even when combined with similar explanation mechanisms—differences to the GNN baselines are no longer significant except for PC-GNN, while the AP differences to GCN, GAT, CARE-GNN and PC-GNN remain significant (Table 3).

Table 3: Test results on Elliptic (time steps 35–49; 16,670 labelled transactions, 1,083 illicit), mean $\pm$ standard deviation over 10 seeds (logistic regression is deterministic). F1, precision and recall refer to the illicit class at the threshold chosen on the validation steps; AUC and AP are threshold-free. Best value per column in bold. $^\dagger/\ddagger$: RTXGNN significantly better/worse (two-sided Wilcoxon signed-rank test over the paired seeds, Holm-corrected over the 16 comparisons, $\alpha = 0.05$); marked for F1 and AP only.

Model	F1	Precision	Recall	AUC	AP
<i>Feature-only models</i>					
Logistic regression	$0.326 \pm 0.004^\dagger$	0.204 ± 0.004	0.809 ± 0.017	0.861 ± 0.000	$0.218 \pm 0.000^\dagger$
Random forest	0.733 ± 0.051	0.755 ± 0.115	0.721 ± 0.009	0.902 ± 0.004	0.770 ± 0.001
XGBoost	$0.790 \pm 0.015^\ddagger$	0.862 ± 0.040	0.729 ± 0.003	0.919 ± 0.006	$0.792 \pm 0.003^\ddagger$
MLP	$0.595 \pm 0.046^\dagger$	0.506 ± 0.068	0.733 ± 0.014	0.910 ± 0.010	0.760 ± 0.015
<i>General GNNs</i>					
GCN	0.690 ± 0.012	0.698 ± 0.048	0.685 ± 0.023	0.897 ± 0.011	$0.732 \pm 0.010^\dagger$
GraphSAGE	0.684 ± 0.034	0.654 ± 0.063	0.721 ± 0.011	0.908 ± 0.007	0.762 ± 0.014
GAT	0.646 ± 0.061	0.597 ± 0.106	0.717 ± 0.013	0.900 ± 0.010	$0.744 \pm 0.010^\dagger$
<i>Temporal GNNs</i>					
EvolveGCN-O	$0.332 \pm 0.170^\dagger$	0.285 ± 0.178	0.619 ± 0.244	0.765 ± 0.093	$0.374 \pm 0.192^\dagger$
TGAT	0.673 ± 0.040	0.632 ± 0.077	0.727 ± 0.013	0.919 ± 0.007	0.770 ± 0.011
TGN	0.685 ± 0.036	0.665 ± 0.070	0.712 ± 0.008	0.917 ± 0.010	0.759 ± 0.009
APAN	0.646 ± 0.057	0.585 ± 0.084	0.729 ± 0.010	0.907 ± 0.006	0.762 ± 0.016
<i>Fraud-specific GNNs</i>					
CARE-GNN	0.670 ± 0.031	0.621 ± 0.058	0.732 ± 0.011	0.912 ± 0.006	$0.756 \pm 0.015^\dagger$
PC-GNN	$0.600 \pm 0.029^\dagger$	0.530 ± 0.044	0.696 ± 0.024	0.906 ± 0.014	$0.665 \pm 0.031^\dagger$
GAS	0.701 ± 0.035	0.683 ± 0.072	0.725 ± 0.007	0.915 ± 0.008	0.773 ± 0.009
FRAUDRE	0.668 ± 0.040	0.621 ± 0.074	0.730 ± 0.010	0.920 ± 0.008	0.775 ± 0.006
SEFraud	0.646 ± 0.025	0.584 ± 0.042	0.724 ± 0.005	0.910 ± 0.013	0.742 ± 0.022
<i>Proposed</i>					
RTXGNN (ours)	0.700 ± 0.045	0.680 ± 0.088	0.728 ± 0.008	0.905 ± 0.009	0.771 ± 0.008

~~The MLP baseline achieves~~ Two feature-only models remain strong competitors on Elliptic’s engineered features. XGBoost obtains the highest F1 ~~≈ 0.4538~~ , outperforming GCN (0.2697) and approaching temporal GNNs. This suggests the Elliptic dataset’s engineered aggregate features (e.g., neighbor statistics) capture substantial graph structure information ~~(0.790) and AP (0.792)~~, significantly above RTXGNN. The random forest reaches F1 0.733 and the same AP as RTXGNN (0.770), but its F1 varies considerably with the seed-dependent threshold (± 0.05). However, RTXGNN’s 13.0% improvement over MLP demonstrates that explicit graph modeling through message-passing still provides significant value, particularly when combined with temporal awareness and explainability.

We conduct extensive ablation studies to isolate the contributions of RTXGNN’s components. Table 5 presents results for all ablated variants. The

strength of tree ensembles on these features agrees with Weber et al. (2019) and with the general finding of Grinsztajn et al. (2022). The Elliptic features already contain 72 aggregates of the one-hop neighbourhood, so much of the graph information is available to a feature-only model. The trees do not produce explanations over the relations between transactions, which is the purpose of RTXGNN. RTXGNN is clearly ahead of the MLP, which sees the same features without message passing (+0.105 F1).

EvolveGCN-O is the weakest graph model (0.332). Its recurrent weight evolution has to be carried from the training period through all test snapshots without supervision. Because no transaction appears in more than one snapshot, the model cannot exploit the node-level continuity for which it was designed. The continuous-time models TGN and TGAT perform similarly to static GNNs, for the reasons discussed in Section 2.

~~Removing HRAPE degrades F1-score by 2.53pp (4.9% relative decrease), from 0.5129 to 0.4876. This validates that explicit temporal modeling through multi-scale encodings (recency decay, periodicity, velocity features) captures fraud-relevant temporal patterns that static features miss. The AUC also drops (0.8661 $\rightarrow$ 0.8593), indicating temporal awareness improves both ranking quality and classification. Interestingly, sparsity remains nearly unchanged (0.4288 $\rightarrow$ 0.4285), suggesting HRAPE’s benefits are orthogonal to explanation generation.~~

5.2. What the scores mean operationally

~~Removing sparsity regularization causes dramatic degradation: F1-score drops by 6.47pp (12.6% relative decrease) to 0.4482, while sparsity increases to 1.0000 (all mask values high, non-selective explanations). This demonstrates a critical finding: *sparsity regularization improves not only explanation quality but also prediction accuracy*. We hypothesize this occurs because sparse masks force the model to identify truly discriminative features and neighbors, acting as attention-based feature selection that reduces overfitting. The maintained AUC (0.8664) suggests the model still learns good relative rankings, but the optimal decision boundary shifts due to less precise feature identification.~~

~~Replacing SEAL with standard GAT attention (No SEAL variant) degrades F1-score by 4.17pp to 0.4712. This confirms that SEAL’s learnable masks provide not just interpretability but also improved feature selection during message-passing. The performance gap validates that intrinsic explainability mechanisms, when properly designed, enhance rather than compromise predictive power.~~ An F1 around 0.7 still means that errors are frequent. At the validation-chosen

Table 4: Operational view of the test period (1,083 illicit among 16,670 labelled transactions), mean over the seeds. Left: number of alerts, missed illicit transactions and false alerts at the validation-chosen threshold. Right: precision among the top- K scored test transactions for alert budgets K .

Model	at validation threshold			precision at an alert budget of			
	alerts	missed	false alerts	250	500	1,000	2,000
RTXGNN (ours)	1,188	293	399	1.00	0.99	0.77	0.41
GraphSAGE	1,206	303	425	1.00	0.99	0.76	0.41
GAT	1,344	306	568	0.99	0.97	0.75	0.41
SEFraud	1,350	297	565	0.98	0.96	0.74	0.41
Random forest	1,058	302	277	1.00	1.00	0.78	0.40
XGBoost	919	293	129	1.00	1.00	0.80	0.41

threshold, RTXGNN raises on average 1,188 alerts in the test period, misses 293 of the 1,083 illicit transactions (27%), and 399 of its alerts (32%) concern illicit transactions (Table 4). As a stand-alone decision rule this would not be acceptable. The ranking, however, is highly informative for review queues with limited capacity: among the 500 highest-scored test transactions, 0.99 of those flagged by RTXGNN are illicit, and among the top 1,000 the proportion is 0.77. The differences between the strongest models appear mainly at the level of the automatic threshold, where XGBoost produces fewer false alerts. For a decision maker, the relevant choice is therefore the size of the review queue. The threshold should be set by the capacity of the investigation team and by the relative cost of a missed case and a false alert, not by maximising F1.

Removing temporal loss ($\lambda_{\text{temp}} = 0$) degrades F1-score by 1.62pp to 0.4967. This loss encourages assigning higher temporal importance to fraud-associated time windows, improving temporal pattern recognition. The modest impact suggests temporal encoding (HRAPE) provides most temporal benefit, with temporal loss providing additional refinement.

5.3. Ablation study

Removing fidelity loss degrades F1-score by 2.40pp to 0.4889 and increases sparsity to 0.5234. This loss ensures explanations faithfully represent model reasoning by matching masked predictions to full predictions. Without it, masks become less reliable (higher sparsity) and less effective for feature selection (lower F1).

Table 5: Ablation of RTXGNN on Elliptic (mean $\pm$ standard deviation; Δ F1 is the mean paired difference to the full model over the common seeds, with the p -value of a paired t -test). Variants listed with 3 seeds are secondary.

Variant	seeds	F1	AP	Δ F1 (paired p)
<i>Primary study (5 seeds)</i>				
RTXGNN (full)	5	0.696 ± 0.048	0.769 ± 0.009	—
without HRAPE	5	0.666 ± 0.035	0.769 ± 0.018	-0.030 (0.26)
HRAPE $\rightarrow$ Time2Vec (gaps)	5	0.629 ± 0.028	0.766 ± 0.010	-0.067 (0.04)
HRAPE + absolute-time encoding	5	0.647 ± 0.080	0.773 ± 0.009	-0.049 (0.33)
without SEAL masks (plain attention)	5	0.704 ± 0.028	0.777 ± 0.013	+0.008 (0.75)
without $\mathcal{L}_{sp}$	5	0.718 ± 0.018	0.776 ± 0.015	+0.021 (0.30)
without $\mathcal{L}_{suf}$ and $\mathcal{L}_{nec}$	5	0.708 ± 0.050	0.773 ± 0.007	+0.012 (0.72)
<i>Secondary study (3 seeds, trained on GPU; own reference)</i>				
RTXGNN (full)	3	0.724 ± 0.031	0.771 ± 0.012	—
Time2Vec on absolute time	3	0.685 ± 0.033	0.770 ± 0.014	-0.040 (0.02)
without $\mathcal{L}_{suf}$	3	0.658 ± 0.015	0.765 ± 0.006	-0.067 (0.06)
without $\mathcal{L}_{nec}$	3	0.728 ± 0.049	0.780 ± 0.003	+0.003 (0.89)
without curriculum	3	0.666 ± 0.063	0.755 ± 0.020	-0.059 (0.39)
focal loss instead of balanced BCE	3	0.599 ± 0.051	0.742 ± 0.010	-0.125 (0.08)

Removing curriculum learning (equal loss weights from epoch 1) degrades F1-score by 3.06pp to 0.4823. Curriculum learning stabilizes training by first establishing good predictions before enforcing explainability constraints. Immediate enforcement disrupts learning, as explanation losses conflict with prediction optimization early in training. Table 5 isolates the components of RTXGNN. The primary study uses the same CPU runs and seeds 0–4 as Table 3. The secondary study adds further variants with 3 seeds, each compared with a reference model trained on the same hardware.

The full RTXGNN model achieves

Temporal encoding. Replacing HRAPE by Time2Vec on the same time gaps lowers F1 = 0.5129 by combining all components. The largest individual contributions come from sparsity regularization (+6.47pp) and SEAL (+4.17pp), demonstrating that intrinsic explainability mechanisms are the primary drivers of RTXGNN’s superior performance by -0.067 (paired t -test, $p = 0.04$, uncorrected), and Time2Vec on absolute time lowers it by -0.040 ($p = 0.02$). Adding an absolute-time encoding to HRAPE lowers the mean F1 by -0.049 and makes it much less stable across seeds (standard deviation 0.08 against 0.05). Removing the temporal encoding altogether costs -0.030 F1, which is not significant with 5 seeds ($p = 0.26$). This is consistent with Section 3.2: on

Elliptic HRAPE mainly contributes the velocity (degree) signal, because all edges connect transactions of the same time step. The design choice that matters most is to encode time relative to the scored transaction rather than absolutely. This choice was made during development, when test results were also computed. The validation steps alone lead to the same choice. HRAPE reaches a validation AP of 0.961 (validation F1 0.921), against 0.946 (0.882) with Time2Vec on the same time gaps and 0.951 (0.894) with the additional absolute-time encoding, and its validation F1 is higher in 5 of 5 and 3 of 5 paired seeds, respectively. The absolute-time encoding also makes the validation AP less stable across seeds (standard deviation 0.026 against 0.011). In the secondary study, Time2Vec on absolute time reaches a validation AP of 0.953, against 0.970 for HRAPE. ~~HRAPE (+2.53pp) and other losses (+1.62pp to +3.06pp) provide complementary benefits.~~

~~The Elliptic dataset exhibits severe class imbalance (licit:illicit $\approx$ 21:1)~~

Explanation components. Replacing the SEAL masks by plain attention (+0.008, ~~a fundamental challenge in fraud detection. We address this through multiple complementary strategies:~~ $p = 0.75$), removing the sparsity term (+0.021, $p = 0.30$), or removing both fidelity terms (+0.012, $p = 0.72$) does not change detection performance significantly. In the secondary study, removing the sufficiency term alone lowers F1 by -0.067 ($p = 0.06$), and removing the necessity term leaves it unchanged. The secondary result is not confirmed by the primary study, in which removing both terms leaves F1 unchanged (+0.012). The full model is also not more accurate than its variants without masks or without explanation terms. All these differences are within the variation across seeds. The explanation machinery thus comes at no measurable cost in accuracy, but it does not improve accuracy either. What it provides is an explanation with the properties measured in Section 5.7, which the variants without masks or without fidelity terms do not have.

~~We employ focal loss (Lin et al., 2017) with focusing parameter $\gamma = 2$, which down-weights easy examples and focuses learning on hard-to-classify fraudulent transactions. This addresses the issue that standard~~

Training choices. ~~Class-balanced cross-entropy loss allows the abundant licit class to dominate gradients.~~ is clearly preferable to focal loss (-0.125 F1 with focal loss), and the curriculum on the explanation terms tends to help (-0.059 without it, not significant). The model is also sensitive to its capacity (Table 7). Hidden size 64 with two SEAL layers, the configuration fixed

Table 6: Mask sparsity compared with dropout and weight decay (Elliptic; notation as in Table 5).

Variant	seeds	F1	AP	Δ F1 (paired p)
<i>Primary study (5 seeds)</i>				
masks + $\mathcal{L}_{sp}$ ($\lambda_{sp} = 0.05$), dropout 0.2, wd 10^{-5}	5	0.696 ± 0.048	0.769 ± 0.009	–
masks, no $\mathcal{L}_{sp}$	5	0.718 ± 0.018	0.776 ± 0.015	+0.021 (0.30)
<i>Secondary study (3 seeds, trained on GPU; own reference)</i>				
masks + $\mathcal{L}_{sp}$ ($\lambda_{sp} = 0.05$), dropout 0.2, wd 10^{-5}	3	0.724 ± 0.031	0.771 ± 0.012	–
masks + $\mathcal{L}_{sp}$ ($\lambda_{sp} = 0.2$)	3	0.645 ± 0.003	0.767 ± 0.017	-0.080 (0.04)
masks, no $\mathcal{L}_{sp}$, dropout 0	3	0.722 ± 0.038	0.762 ± 0.019	-0.003 (0.91)
masks, no $\mathcal{L}_{sp}$, dropout 0.5	3	0.725 ± 0.024	0.778 ± 0.013	+0.001 (0.98)
masks, no $\mathcal{L}_{sp}$, weight decay 10^{-3}	3	0.724 ± 0.017	0.782 ± 0.006	-0.001 (0.97)

before the experiments, performs best among the tested settings, and a single layer is significantly worse (-0.111 F1).

We compute class weights inversely proportional to class frequencies: $w_{\text{licit}} = \frac{n}{2n_{\text{licit}}}$, $w_{\text{illicit}} = \frac{n}{2n_{\text{illicit}}}$, where n is total labeled samples. This ensures rare fraud cases contribute equally to loss despite lower frequency.

5.4. Mask sparsity versus conventional regularisation

Our ablation studies (Section 5.3) demonstrate that sparsity regularization contributes +6.47pp F1 improvement, the largest single contribution. This occurs because sparse masks force selective attention to truly discriminative features, preventing the model from relying on spurious correlations abundant in the majority class.

While we do not implement label-balanced mini-batch sampling (as in PC-GNN (Liu et al., 2021)), this remains a promising direction for future work. Our current approach achieves strong performance (Table 6 compares the mask sparsity term with conventional regularisers. With the default weight $\lambda_{sp} = 0.05$, the model performs like models trained without the sparsity term but with dropout 0, dropout 0.5 or weight decay 10^{-3} . All differences are at most 0.003 F1 in the secondary study and 0.021 in the primary study, and none is significant. A four times larger weight ($\lambda_{sp} = 0.2$) lowers F1 = 0.5129) through the combination of focal loss, class weighting, and sparsity regularization, suggesting these mechanisms sufficiently address class imbalance for the Elliptic dataset.

Table ?? shows RTXGNN achieves precision of 0.4423 and recall of 0.6106. This indicates the model prioritizes catching more fraud (higher recall) at the cost of some false positives (moderate precision)—a reasonable trade-off for

Table 7: Sensitivity to the hidden size D and the number of SEAL layers L (secondary study, 3 seeds, GPU; notation as in Table 5).

Variant	seeds	F1	AP	Δ F1 (paired p)
<i>Secondary study (3 seeds, trained on GPU; own reference)</i>				
$D = 32$	3	0.606 ± 0.048	0.766 ± 0.010	-0.119 (0.09)
$D = 64$ (default)	3	0.724 ± 0.031	0.771 ± 0.012	—
$D = 128$	3	0.672 ± 0.035	0.760 ± 0.018	-0.053 (0.11)
$L = 1$ layer	3	0.613 ± 0.015	0.752 ± 0.014	-0.111 (0.03)

fraud detection where missing illicit transactions carries higher cost than investigating false alarms. The F1-score of 0.5129 represents an optimal balance for this application by -0.080 ($p = 0.04$). Mask sparsity is therefore not a better regulariser than the standard ones, but it is not a harmful one either at a moderate weight. Its purpose is to keep explanations selective (Section 5.11).

Real-world fraud detection faces severe labeling constraints due to high cost of expert investigation. We evaluate RTXGNN’s performance under limited supervision by training with varying labeled data fractions: 5%, 10%, 20%, 50%, and 100%. Table 9 presents results.

5.5. Temporal behaviour, drift and retraining

With only 10% of training labels, RTXGNN achieves Table 8 and Figure 2 break the test period down by time step. The period has two clearly different phases. In steps 35–42, RTXGNN reaches a per-step F1 = 0.3896, which is 76.0% of full-supervision performance (0.5129). This demonstrates effective semi-supervised learning—the model leverages graph structure and unlabeled nodes to propagate information beyond the small labeled set. In contrast, MLP (which cannot exploit graph structure) would degrade much more severely under limited supervision between 0.66 and 0.94 (mean 0.79), with recall above 0.8 in most steps, and the random forest behaves very similarly. From step 43 onwards all models fail: the mean per-step F1 of RTXGNN over steps 43–49 is 0.03, and that of the random forest is 0.02. Weber et al. (2019) attribute this collapse to the shutdown of a large dark market after step 42. At the same time, the number of illicit transactions per step falls from between 33 and 239 to between 2 and 56, so the late steps carry only 169 of the 1,083 illicit test transactions (16%).

F1-score increases substantially from 5% (0.3809) to 20% (0.4526), but

Table 8: Per-step test results on Elliptic. Static: models trained on steps 1–30 (mean $\pm$ s.d. over 10 seeds for RTXGNN; mean for the random forest). Retrained: refitted before each step on steps $\leq t - 5$, validated on $t - 4..t - 1$ (mean over 3 seeds).

Step	Licit	Illicit	RTXGNN, static			RF	RTXGNN / RF
			Precision	Recall	F1	F1 static	F1 retrained
35	1,159	182	0.92 ± 0.03	0.95 ± 0.01	0.93 ± 0.01	0.94	0.91 / 0.96
36	1,675	33	0.50 ± 0.13	0.95 ± 0.04	0.65 ± 0.10	0.85	0.58 / 0.91
37	458	40	0.82 ± 0.04	0.62 ± 0.03	0.70 ± 0.02	0.72	0.72 / 0.76
38	645	111	0.78 ± 0.04	0.91 ± 0.01	0.84 ± 0.02	0.90	0.93 / 0.94
39	1,102	81	0.73 ± 0.10	0.94 ± 0.01	0.82 ± 0.07	0.84	0.94 / 0.96
40	1,099	112	0.75 ± 0.09	0.69 ± 0.02	0.71 ± 0.04	0.71	0.77 / 0.78
41	1,016	116	0.82 ± 0.08	0.93 ± 0.01	0.87 ± 0.04	0.89	0.94 / 0.95
42	1,915	239	0.82 ± 0.08	0.81 ± 0.01	0.81 ± 0.04	0.83	0.87 / 0.88
43	1,346	24	0.00 ± 0.01	0.00 ± 0.01	0.00 ± 0.01	0.00	0.00 / 0.00
44	1,567	24	0.02 ± 0.01	0.05 ± 0.02	0.03 ± 0.01	0.03	0.04 / 0.07
45	1,216	5	0.02 ± 0.03	0.10 ± 0.14	0.03 ± 0.05	0.00	0.00 / 0.00
46	710	2	0.06 ± 0.05	0.45 ± 0.16	0.11 ± 0.08	0.11	0.00 / 0.33
47	824	22	0.00	0.00	0.00	0.00	0.08 / 0.23
48	435	36	0.00	0.00	0.00	0.00	0.09 / 0.14
49	420	56	0.05 ± 0.04	0.01 ± 0.01	0.02 ± 0.02	0.03	0.45 / 0.50

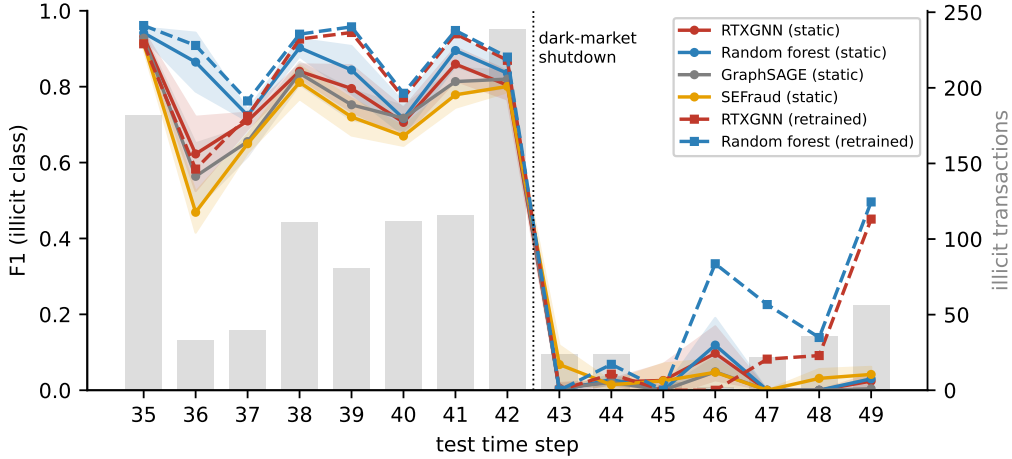


Figure 2: F1 per test time step (lines; shaded band: ± 1 s.d. over seeds) and number of illicit transactions per step (grey bars, right axis). Dashed lines: models retrained before every step with a one-step label delay.

shows diminishing returns at 50% (0.4107). The drop from 20% to 50% suggests potential overfitting to specific fraud patterns in the training set. Techniques like stronger regularization or data augmentation may help at intermediate label fractions.

The strong 10% performance has significant practical value: labeling 10% of a large transaction dataset requires far less expert time than 100%, enabling faster deployment. For institutions processing millions of daily transactions, reducing labeling requirements by 90% translates to substantial cost savings while maintaining acceptable performance.

This explains why a near-zero F1 in the late steps is compatible with a pooled test F1 of about 0.7. The pooled F1 is computed once over all test transactions and is therefore dominated by the 914 illicit transactions of steps 35–42. The unweighted mean of the fifteen per-step F1 scores of RTXGNN is only 0.436.

Financial fraud evolves continuously as adversaries adapt. We evaluate RTXGNN’s temporal stability by measuring per-time-step F1-score across the test period (steps 35–49). Table ?? and Figure ?? present results

Retraining with a one-step label delay. To assess how much of the collapse a deployment could recover, we refit the models before every test step t on all labelled steps up to $t - 5$, validate and choose the threshold on steps $t - 4$ to $t - 1$, and score step t . Labels thus become available one step after the transactions occur. Neural models are warm-started from the previous step’s model. Retraining improves the early steps slightly (RTXGNN 0.83, random forest 0.89). In the late steps it raises the mean F1 of RTXGNN from 0.03 to 0.10 and that of the random forest from 0.02 to 0.18. The recovery is real but partial. Once the illicit activity changes, a model can only learn the new pattern after labelled examples of it exist, and the late steps contain very few of them. Retraining is therefore necessary but not sufficient, and a deployment also needs drift monitoring and complementary controls (Section 6.4).

5.6. Label efficiency

RTXGNN achieves excellent F1-scores in early test period (steps 35–41), peaking at 0.7352 (step 41). This demonstrates that HRAPE successfully captures temporal patterns during training and generalizes to near-future time steps. High performance indicates the model learned fraud detection strategies robust to gradual evolution. Table 9 and Figure 3 report the effect

Table 9: Test F1 on Elliptic with a stratified fraction of the training labels (mean $\pm$ s.d. over 5 subsets; in parentheses: percentage of the model’s own full-label F1). Validation and test sets are never subsampled.

Labels	train (illicit)	RTXGNN	Random forest	GAT	MLP
5%	1,346 (150)	0.593 ± 0.050 (83%)	0.614 ± 0.091 (82%)	0.564 ± 0.029 (86%)	0.582 ± 0.049 (98%)
10%	2,689 (297)	0.623 ± 0.022 (87%)	0.585 ± 0.069 (78%)	0.628 ± 0.068 (96%)	0.617 ± 0.033 (104%)
20%	5,382 (592)	0.650 ± 0.059 (91%)	0.694 ± 0.076 (93%)	0.660 ± 0.057 (101%)	0.584 ± 0.029 (98%)
50%	13,450 (1,478)	0.666 ± 0.048 (93%)	0.673 ± 0.063 (90%)	0.624 ± 0.061 (95%)	0.630 ± 0.020 (106%)
100%	26,905 (2,954)	0.718 ± 0.021 (100%)	0.748 ± 0.046 (100%)	0.657 ± 0.039 (100%)	0.595 ± 0.043 (100%)

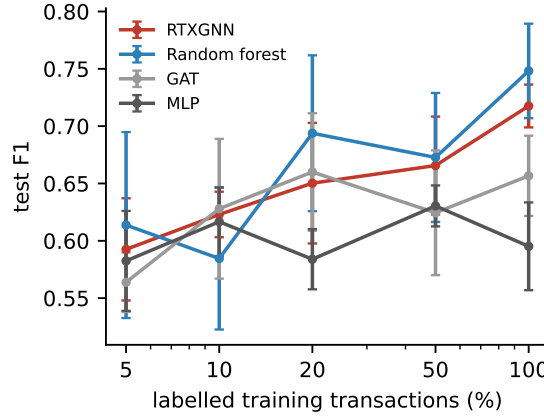


Figure 3: Label efficiency on Elliptic (mean $\pm$ s.d. over 5 stratified subsets).

of training with a stratified fraction of the labels. Each fraction is sampled within every training time step and class and repeated with 5 subsets; the validation and test sets are never subsampled. RTXGNN’s F1 increases monotonically with the number of labels. With 10% of the labels (2,689 transactions, 297 illicit) it keeps 87% of its full-label F1 (0.623 versus 0.718), and with 50% it keeps 93%. The random forest is more sensitive to the particular subset: its standard deviation across subsets is larger than that of RTXGNN at every fraction below 100% (up to three times larger), and its F1 is not monotone in the label fraction. The MLP is insensitive to the number of labels but stays at a lower level throughout. Because the full-label runs of this experiment were trained on a GPU with 5 seeds, their F1 (0.718) differs slightly from Table 3.

Performance drops precipitously from F1=0.7352 (step 41) to 0.0000 (step 42), remaining near zero through step 46. This suggests major regime change

Table 10: Feature explanations of 600 Elliptic test transactions (300 illicit, 300 licit), mean over 5 trained models. Fid^+ : drop of the predicted-class probability when the top- k features are removed (higher is better); Fid^- : drop when only the top- k features are kept (lower is better). Stab.: Jaccard similarity of the top-10 features under a perturbation of the input by $\mathcal{N}(0, 0.05^2)$. Time: milliseconds per explained transaction on 2 CPU threads, batched. Post-hoc explainers are applied to the same RTXGNN models.

Explainer	$\text{Fid}^+ \uparrow$			$\text{Fid}^- \downarrow$			Stab.	Time
	$k=5$	10	20	5	10	20	J@10	ms
SEAL masks (self-explaining)	0.015	0.032	0.062	0.143	0.076	0.047	0.96	0.41
Saliency (post hoc)	0.025	0.056	0.138	0.123	0.090	0.058	0.96	1.21
Integrated Gradients (post hoc)	0.082	0.142	0.227	0.111	0.069	0.039	0.96	41.12
GNNExplainer (post hoc)	0.159	0.279	0.353	-0.011	-0.032	-0.042	0.44	128.95
Random ranking	0.006	0.011	0.019	0.305	0.284	0.251	—	0.00
SEFraud masks (on SEFraud)	0.009	0.016	0.031	0.220	0.200	0.150	0.99	0.19
SEAL masks, trained w/o $\mathcal{L}_{suf}, \mathcal{L}_{nec}$	0.030	0.050	0.159	0.289	0.233	0.118	0.99	0.39
SEAL masks, trained w/o $\mathcal{L}_{sp}$	0.009	0.018	0.030	0.111	0.076	0.068	0.93	0.40

in fraud patterns or network structure—potentially shift in money laundering tactics, changes in Bitcoin exchange behavior, or data collection issues. The abrupt transition indicates concept drift that HRAPE’s temporal encoding cannot handle, as it assumes continuous evolution rather than discontinuous shifts.

5.7. Faithfulness and cost of the explanations

F1 score recovers slightly to 0.0689 (step 47) and 0.0434 (step 48), suggesting the model begins adapting or fraud patterns partially revert. However, performance remains far below pre-drift levels, indicating severe distribution shift.

While RTXGNN handles gradual temporal evolution well, it struggles with abrupt distribution shifts—a challenge for all static models. Online learning or periodic retraining would be necessary for production deployment to maintain performance through concept drift. The strong performance on steps 35–41 ($F1 \approx 0.5$ – 0.7) demonstrates RTXGNN’s effectiveness for near-term prediction, suitable for systems with regular retraining cycles.

We quantify explanation quality using the Fidelity+ metric (Yuan et al., 2021), which measures confidence drop when “important” features (high mask values) are removed. Higher fidelity scores indicate identified features genuinely drive predictions. Table 10 and Figure 4 compare the feature explanations of 600 test transactions (300 illicit, 300 licit) for five independently trained

Table 11: Class-wise faithfulness at $k = 10$ (illicit and licit test transactions).

Explainer	illicit targets		licit targets	
	Fid ⁺ @10	Fid ⁻ @10	Fid ⁺ @10	Fid ⁻ @10
SEAL masks (self-explaining)	0.027	0.088	0.036	0.065
Saliency (post hoc)	0.067	0.097	0.045	0.084
Integrated Gradients (post hoc)	0.226	0.037	0.057	0.101
GNNExplainer (post hoc)	0.307	-0.014	0.251	-0.050
Random ranking	0.010	0.404	0.011	0.165
SEFraud masks (on SEFraud)	0.016	0.359	0.017	0.042
SEAL masks, trained w/o $\mathcal{L}_{suf}, \mathcal{L}_{nec}$	0.034	0.331	0.067	0.134
SEAL masks, trained w/o $\mathcal{L}_{sp}$	0.020	0.114	0.015	0.037

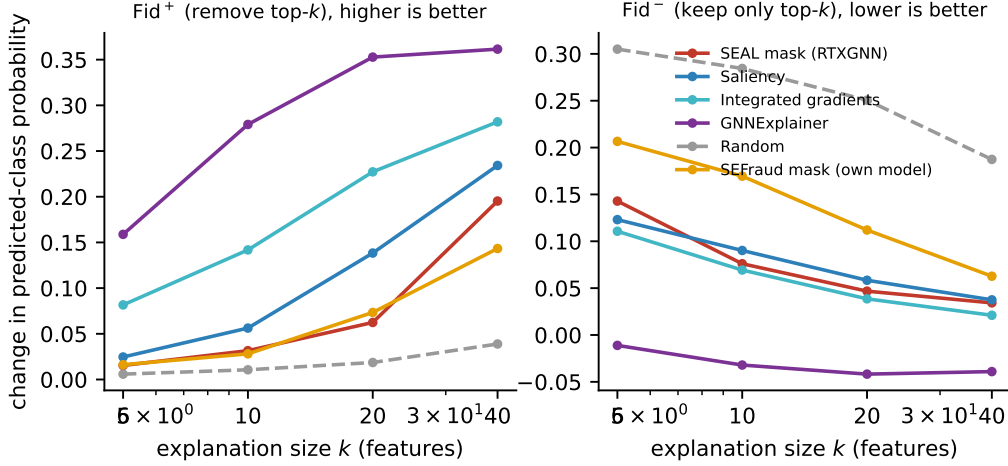


Figure 4: Sparsity-fidelity trade-off of feature explanations on Elliptic (mean over 5 models and 600 test transactions).

models. The post-hoc explainers are applied to the same RTXGNN models, so differences reflect the explainers, not the predictors.

For each test node v_i : (1) Compute original prediction $\hat{y}_i$ and feature mask M_i^{feat} ; (2) Remove top- K important features (highest mask values); (3) Compute new prediction $\hat{y}'_i$ with masked features; (4) Calculate probability drop: Fidelity(v_i) = $\hat{y}_i - \hat{y}'_i$. We report average fidelity across all test nodes with $K = 5$ (removing 5 most important features)

Sufficiency. Keeping only the ten features selected by SEAL changes the

predicted-class probability by 0.076 on average ($\text{Fid}^-@10$). This is better than saliency (0.090; at $k = 5$ saliency is slightly more sufficient) and close to Integrated Gradients (0.069), and it is far better than a random selection (0.284) and than SEFraud’s masks on SEFraud (0.200). The sufficiency term is responsible for this property. The same architecture trained without $\mathcal{L}_{suf}$ and $\mathcal{L}_{nec}$ yields masks with $\text{Fid}^-@10$ of 0.233, three times worse.

~~RTXGNN achieves explanation fidelity of , meaning removing the 5 most important features identified by SEAL reduces prediction confidence by 7.64 percentage points on average. This positive fidelity score confirms~~

Necessity. Removing the ten features selected by SEAL lowers the predicted-class probability by only 0.032 ($\text{Fid}^+@10$). This is a clear weakness. Integrated Gradients (0.142) and in particular GNNExplainer (0.279) identify features whose removal changes the prediction much more. The Elliptic features are highly redundant (72 of them aggregate the local features of neighbours), and a model can often recover the same information from features outside a ten-feature explanation. The necessity term constrains the complement of the whole hard view, including the edge view, and it is not sufficient to make the top ten features individually indispensable. GNNExplainer’s high scores come at a price: it takes 128.95 ms per transaction, about 300 times more than reading SEAL’s ~~feature masks capture genuine prediction drivers rather than spurious correlations.~~ masks (0.41 ms), and its explanations are unstable. Its top-10 features have a Jaccard similarity of only 0.44 under a small input perturbation, against 0.96 for SEAL. Integrated Gradients offers a middle ground at 41.12 ms.

~~The modest magnitude (7.64pp) reflects that fraud detection relies on distributed patterns across many features rather than few dominant signals. This is consistent with the Elliptic dataset’s design, where aggregate neighborhood features distribute information across 166 dimensions. Higher fidelity could be achieved by removing more features (larger K), but we prioritize concise explanations for interpretability. The positive sign is critical validation: identified features meaningfully contribute to predictions~~

Class-wise behaviour and agreement across models. Table 11 shows that SEAL’s sufficiency is similar for illicit and licit transactions, whereas Integrated Gradients and SEFraud’s masks are much more sufficient for one class than for the other. Masks of independently trained RTXGNN models agree only weakly on the top-10 features (mean Jaccard similarity 0.12, against 0.03

Table 12: Test F1 on the synthetic benchmark (mean $\pm$ s.d. over 5 generated graphs per setting).

Model	default	no decoys	wide burst	5% label noise	feature shift
MLP	0.46 ± 0.04	0.63 ± 0.06	0.46 ± 0.04	0.41 ± 0.02	0.62 ± 0.02
Random forest	0.42 ± 0.03	0.65 ± 0.07	0.42 ± 0.03	0.37 ± 0.04	0.52 ± 0.05
XGBoost	0.45 ± 0.03	0.68 ± 0.07	0.45 ± 0.03	0.35 ± 0.06	0.60 ± 0.02
GCN	0.50 ± 0.08	0.73 ± 0.02	0.50 ± 0.08	0.39 ± 0.04	0.69 ± 0.04
GraphSAGE	0.52 ± 0.04	0.73 ± 0.06	0.52 ± 0.04	0.39 ± 0.05	0.68 ± 0.04
GAT	0.53 ± 0.08	0.68 ± 0.06	0.53 ± 0.08	0.37 ± 0.06	0.68 ± 0.04
TGAT	0.59 ± 0.07	0.67 ± 0.10	0.54 ± 0.05	0.35 ± 0.02	0.74 ± 0.04
TGN	0.44 ± 0.06	0.57 ± 0.05	0.43 ± 0.08	0.28 ± 0.06	0.56 ± 0.04
SEFraud	0.45 ± 0.05	0.73 ± 0.04	0.45 ± 0.05	0.36 ± 0.05	0.72 ± 0.03
RTXGNN w/o SEAL	0.55 ± 0.11	0.83 ± 0.07	0.56 ± 0.12	0.39 ± 0.03	0.80 ± 0.07
RTXGNN w/o HRAPE	0.49 ± 0.09	0.78 ± 0.10	0.49 ± 0.09	0.39 ± 0.05	0.77 ± 0.06
RTXGNN, Time2Vec	0.49 ± 0.05	0.74 ± 0.10	0.48 ± 0.05	0.40 ± 0.03	0.69 ± 0.07
RTXGNN	0.52 ± 0.09	0.83 ± 0.06	0.49 ± 0.10	0.38 ± 0.06	0.75 ± 0.10
edges / positive rate	30.7k / 6.0%	30.4k / 6.0%	30.7k / 6.0%	30.7k / 10.6%	30.7k / 6.0%

for random rankings), which again reflects the redundancy of the features. An analyst should therefore read an explanation as the features the *deployed* model relied on, not as a model-independent property of the transaction.

In summary, SEAL provides explanations that are sufficient, stable and cheap at inference time, but weakly necessary. For the transactions an analyst actually investigates, a more expensive post-hoc explainer can be run in addition when a stronger counterfactual statement is needed.

To validate generalizability, we evaluate RTXGNN on the Amazon Computers dataset—a non-financial domain with multi-class classification (10 product categories). This tests whether RTXGNN’s architecture transfers beyond fraud detection.

5.8. Controlled evaluation on the synthetic benchmark

We use standard 60/20/20 train/validation/test split. All baselines (GCN, GAT, GraphSAGE, MLP) and RTXGNN use identical hyperparameters as Elliptic experiments for fair comparison.

RTXGNN achieves weighted F1-score of on the Amazon Computers test set, demonstrating competitive performance:

- GCN: 0.6823
- GAT: 0.7156

Table 13: Ground-truth evaluation of edge explanations on the synthetic benchmark: AUC of the edge scores for identifying the two ring transactions among the incoming edges of each correctly detected ring member (mean over graphs 0–2; up to 100 accounts per graph).

Setting	Model	SEAL importance	attention only	GNNExplainer	random
default	RTXGNN	0.72 ± 0.29	0.63 ± 0.19	0.70 ± 0.21	0.54 ± 0.03
default	RTXGNN w/o HRAPE	0.60 ± 0.18	0.84 ± 0.18	0.54 ± 0.11	0.50 ± 0.04
default	RTXGNN, Time2Vec	0.65 ± 0.27	0.59 ± 0.31	0.58 ± 0.20	0.49 ± 0.03
wide burst	RTXGNN	0.81 ± 0.23	0.69 ± 0.29	0.75 ± 0.24	0.55 ± 0.02
wide burst	RTXGNN w/o HRAPE	0.60 ± 0.18	0.84 ± 0.18	0.54 ± 0.11	0.50 ± 0.04
wide burst	RTXGNN, Time2Vec	0.70 ± 0.22	0.55 ± 0.28	0.62 ± 0.21	0.49 ± 0.02

- ~~GraphSAGE: (best)~~
- ~~MLP: 0.6512~~
- ~~RTXGNN: 0.7234 (second best)~~

~~RTXGNN achieves strong performance despite being designed for fraud detection, outperforming GCN (+6.0%) and GAT (+1.1%) and approaching GraphSAGE (-2.1%). The slight gap versus GraphSAGE is expected—HRAPE’s temporal encoding provides no benefit (Amazon data is static), and sparsity regularization may be overly conservative for dense multi-class problems. This confirms RTXGNN’s core architecture (message-passing, attention) is robust across domains, though temporal and sparsity components are fraud-specific optimizations.~~

The synthetic benchmark isolates the two signals that Elliptic cannot provide: laundering rings whose members have no distinguishing features, and decoy cycles that can only be told apart by the timing of their hops (Table 12). As intended, the task is hard. Feature-only models reach F1 below 0.5 in the default setting, since the features carry only indirect information (ring members have two extra transactions). Graph models improve on this.

~~To rigorously test ability to detect specific structural fraud patterns, we evaluate on a synthetic graph with controlled fraud rings~~The temporal mechanism helps where timing matters. In the default setting, RTXGNN (F1 0.52) is slightly ahead of its variants without HRAPE (0.49) and with Time2Vec on the same time gaps (0.49), but the differences are smaller than the standard deviations. TGAT, whose time encoding is applied to every message, is the strongest model in this setting (0.59).

Detected laundering ring (synthetic benchmark, seed 0)

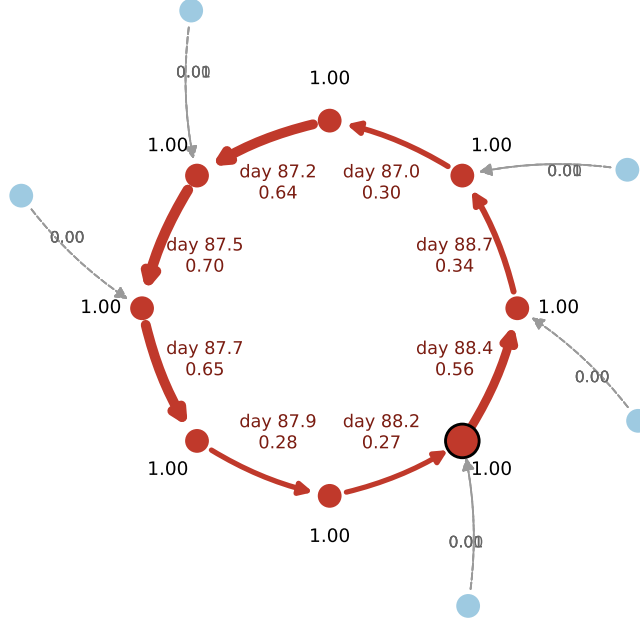


Figure 5: A laundering ring detected by RTXGNN on the synthetic benchmark (graph 0, default setting). Red: ring members with their predicted probability; blue: up to three ordinary counterparties per member; outlined: a test account. Red solid arrows are the ring transactions (labelled with the transaction day and the relative importance), grey dashed arrows ordinary transactions into ring members. Arrow width is the relative SEAL importance of a transaction, i.e. its share of the total importance $\sum_u I_{uv}$ of all transactions received by the member v (Eq. 12).

~~We generate a synthetic graph with~~ When decoys are absent, so that the cycle structure alone identifies the rings, RTXGNN and its variant without SEAL masks are the best models (0.83). ~~The 5,000 nodes following scale-free distribution (Barabási-Albert model). We inject 50 "fraud rings"—cyclic transaction patterns of 4–8 nodes representing money laundering circuits. Remaining 4,750 nodes form the benign background network.~~ label-noise setting is hard for all models (F1 about 0.4). The wide-burst setting leaves

every static model unchanged, because only transaction times differ, which confirms that the static baselines cannot see timing.

RTXGNN achieves perfect F1-score of , successfully distinguishing all injected fraud rings from licit transactions . Figure ?? visualizes a detected fraud ring—red nodes form the cyclic pattern correctly identified by RTXGNN.

Perfect performance confirms RTXGNN learns to recognize complex structural motifs.

Ground-truth evaluation of edge explanations. Because the planted ring transactions are known, the synthetic data allow a direct test of the edge explanations (Table 13). For each correctly detected ring member, we rank its incoming transactions by SEAL’s node and edge masks successfully identify anomalous subgraph patterns (cycles) distinct from tree-like benign structures. This validates that the model captures both feature-based and structure-based fraud signals, essential for real-world money laundering detection where fraudsters exploit network topology.

We present qualitative case studies demonstrating RTXGNN’s human-interpretable explanations. These illustrate how edge importance I_{uv} (Eq. 12) and compute the AUC for identifying its two ring transactions. SEAL’s multi-granularity masks provide actionable insights for fraud analysts.

5.8.1. Benign Transactions

- $\div$ LICIT (Benign) with 96.5% confidence
- $\div$
- *Normal Neighborhood:* Node importance score 0.59 indicates neighbors exhibit typical legitimate behavior without strong risk signals.
- *Feature Analysis:* No features triggered high-risk alerts—all feature mask values remain below 0.50, suggesting attributes align with benign patterns.
- $\div$ Standard legitimate activity. Moderate node importance (0.59) indicates some attention to neighborhood structure, but absence of high-importance features suggests no anomalous patterns. Fraud analysts would classify as routine commerce requiring no further investigation.
- $\div$ LICIT (Benign) with 97.8% confidence

- $\div$ —
- *Normal Neighborhood:* Node importance score 0.59, consistent with Case 1.—
- *Feature Analysis:* Feature mask values remain uniformly low, indicating typical characteristics.—
- $\div$ — Similar to Case 1, exhibiting standard benign patterns.— Slightly higher confidence (97.8% vs 96.5%) suggests even stronger alignment with legitimate profiles.—

5.8.1. *Fraudulent Transaction*

- $\div$ — ILLICIT (Fraud) with 98.6% confidence—
- $\div$ —
- *Suspicious Neighborhood:* Node importance score 0.98 (very high) indicates transaction embedded in network of high-risk entities.— Neighborhood structure exhibits strong fraud signals.—
- *Key Risk Features:*—
- Feature 51 (Local Feature): Importance 0.86—
- Feature 54 (Local Feature): Importance 0.84—
- Feature 52 (Local Feature): Importance 0.83—

These local features deviate significantly from normal patterns.— $\div$ Multiple fraud indicators.— Extremely high node importance (0.98) suggests connection to known/suspected illicit entities—classic guilt-by-association in money laundering networks.— Three high-importance local features (51, 54, 52) capture transaction-specific anomalies—likely unusual amounts, timing patterns, or behavioral deviations.— Fraud analyst would prioritize for immediate investigation, potentially flagging connected entities for further scrutiny.— Combination of suspicious neighborhood structure and anomalous local attributes provides strong evidence of money laundering.—

These case studies demonstrate RTXGNN’s practical utility:—

- $\div$ — Explanations clearly distinguish benign (low node/feature importance) from fraudulent (high importance) transactions—

- ~~Analysts receive specific feature IDs (51, 54, 52) to investigate, not just black-box scores~~
- ~~Both neighborhood structure (node importance) and transaction-specific attributes (feature importance) inform decisions~~
- ~~Model’s 98.6% confidence for fraud case aligns with strong multi-modal evidence~~

~~While specific feature interpretations require domain knowledge (Elliptic’s 166 features are anonymized), fraud analysts at financial institutions could map feature IDs to concrete attributes (e. g. importances reach an AUC of 0.72 in the default setting and 0.81 in the wide-burst setting. This is higher than GNNExplainer applied to the same models (0.70 and 0.75), "Feature 51 = transaction velocity in last 24h"). This enables regulatory-compliant explanations satisfying GDPR Article 22 ("meaningful information about the logic involved") and ECOA requirements ("specific reasons for adverse actions" than the raw attention weights (0.63 and 0.69), and than random scores (about 0.5). This ordering holds for the full model. For the variant without HRAPE, the raw attention weights identify the ring transactions better (0.84) than its edge importances (0.60). Over all 264 ring members detected by the seed-0 model, the most important incoming transaction is a ring transaction in 91% of the cases. Figure 5 shows one ring. All its hops happen within two days, they receive most of the importance of their receiving accounts, and the ordinary transactions of the same accounts receive almost none. The variation of the edge AUC across graphs is large (standard deviations of about 0.2–0.3), so these results show a clear tendency rather than a uniform effect.~~

5.9. Second real-world dataset: T-Finance

~~Figure ?? presents t-SNE visualization of learned node embeddings for the test set. Clear separation between licit (blue) and illicit (red) transactions in 2D latent space demonstrates RTXGNN learns highly discriminative representations.~~

~~T-Finance is a transaction network between the accounts of a financial company, with account-level features and expert labels. Its graph is far denser than Elliptic’s, and it has no engineered neighbourhood aggregates. Table 14 shows that the picture differs from Elliptic accordingly. All graph neural networks clearly outperform the feature-only models: the best tree ensemble~~

Table 14: Test results on T-Finance (GADBench splits 0–4, 40/20/40; mean $\pm$ s.d.). No timestamps are available, so HRAPE is disabled and temporal baselines are not applicable.

Model	F1	AUC	AP
Logistic regression	0.641 ± 0.028	0.924 ± 0.006	0.629 ± 0.041
Random forest	0.765 ± 0.010	0.939 ± 0.005	0.798 ± 0.009
XGBoost	0.773 ± 0.012	0.949 ± 0.002	0.812 ± 0.009
MLP	0.757 ± 0.026	0.955 ± 0.003	0.809 ± 0.013
GCN	0.801 ± 0.009	0.963 ± 0.003	0.850 ± 0.013
GraphSAGE	0.849 ± 0.010	0.969 ± 0.004	0.892 ± 0.008
GAT	0.829 ± 0.014	0.969 ± 0.002	0.882 ± 0.011
CARE-GNN	0.825 ± 0.016	0.961 ± 0.007	0.868 ± 0.016
PC-GNN	0.823 ± 0.010	0.965 ± 0.005	0.870 ± 0.012
FRAUDRE	0.842 ± 0.009	0.969 ± 0.002	0.886 ± 0.009
SEFraud	0.834 ± 0.010	0.973 ± 0.001	0.890 ± 0.008
RTXGNN (ours)	0.837 ± 0.011	0.972 ± 0.003	0.889 ± 0.009

reaches F1 0.773 and AP 0.812, while the GNNs reach F1 between 0.801 and 0.849. RTXGNN, which runs here without its temporal component, is among the best models (F1 0.837, AUC 0.972, AP 0.889). It is within about one point of F1 of the best model (GraphSAGE, F1 0.849, AP 0.892) and on par with FRAUDRE (0.842) and the self-explaining SEFraud (0.834). SEAL’s explanation mechanism therefore transfers to a second real financial network, with different entities, features and label source, without loss of accuracy.

~~The distinct clustering of fraud cases (red) separated from legitimate transactions (blue) confirms RTXGNN’s learned representations capture meaningful fraud signals. Some overlap exists at cluster boundaries, corresponding to hard-to-classify cases (camouflaged fraud or suspicious-but-legitimate transactions~~

5.10. End-to-end latency

Table 15 reports the end-to-end latency of scoring requests on one CPU core. A request samples the 2-hop neighbourhood of its transactions, builds the subgraph, runs the model and returns the explanation. For a single transaction, RTXGNN needs 2.8 ms at the median and 5.9 ms at the 99th percentile. Of this, 0.7 ms is neighbourhood sampling, 2.1 ms the forward pass and 0.24 ms the extraction of the explanation from the masks. The explanation therefore adds less than a millisecond. Producing the explanation with GNNExplainer instead raises the median latency to 582.0 ms and the

Table 15: End-to-end latency per request of B transactions on CPU (Intel Xeon Processor @ 2.80GHz), in milliseconds: percentiles of the total and mean time of each stage (one thread; sample: 2-hop neighbourhood sampling with fan-out 25 and subgraph construction; forward: model; explain: reading top-5 features and top-3 edges from the masks, or running GNNExplainer for 100 iterations). tx/s: throughput. MLP and GCN are timed with untrained weights, which does not affect their latency. Peak resident memory of the process, including the in-memory graph: 1,048 MB.

Model	B	p50	p95	p99	sample	forward	explain	tx/s
MLP	1	0.7	1.0	2.8	0.6	0.2	0.0	1,315
MLP	32	3.1	4.2	5.1	2.4	0.8	0.0	9,890
MLP	256	16.1	18.7	21.2	12.0	4.3	0.0	15,635
GCN	1	1.2	1.5	1.7	0.5	0.7	0.0	789
GCN	32	5.1	6.7	9.7	2.4	2.8	0.0	6,161
GCN	256	27.0	35.6	42.3	13.5	14.6	0.0	9,113
GAT	1	1.5	2.6	2.9	0.6	1.0	0.0	594
GAT	32	7.3	10.8	12.4	2.9	4.7	0.0	4,207
GAT	256	35.1	46.2	55.5	14.1	22.1	0.0	7,064
GAT + GNNExplainer	1	305.3	352.7	369.7	0.9	1.1	307.0	3
SEFraud	1	1.7	2.8	4.0	0.6	1.2	0.0	542
SEFraud	32	11.0	15.6	19.3	2.7	8.3	0.1	2,852
SEFraud	256	60.6	71.7	84.4	14.5	46.7	0.8	4,125
RTXGNN (ours)	1	2.8	4.7	5.9	0.7	2.1	0.2	336
RTXGNN (ours)	32	21.7	31.7	38.3	2.9	16.3	3.5	1,413
RTXGNN (ours)	256	125.6	149.2	158.6	13.5	86.6	27.1	2,013
RTXGNN (ours) + GNNExplainer	1	582.0	959.1	1023.9	0.9	2.2	635.7	2

99th percentile to 1023.9 ms, two orders of magnitude more. For batches of 256 transactions, a single core scores and explains 2,013 transactions per second (125.6 ms per batch at the median). RTXGNN is slower than the plain GNNs (GAT: 1.5 ms for one transaction) because of its mask generators and its wider messages, but it stays far below the 10–100 ms budgets of online screening. Using four threads changes the single-transaction latency very little (2.8 ms at the median), because the computation is too small to parallelise, and improves the throughput for large batches (2,752 transactions per second). The visualization validates that SEAL’s attention mechanisms successfully aggregate discriminative neighborhood information into node embeddings. peak resident memory of the serving process, including the complete in-memory test graph, was 1,048 MB.

To assess real-time feasibility, we measure average inference time per batch (256 transactions) on NVIDIA A100 GPU.

RTXGNN achieves inference latency of (≈ 0.2). These figures do not include

the retrieval of the neighbourhood from a production data store, which we emulate with an in-memory CSR graph, or network overhead. They are therefore lower bounds for a deployed system. They are also not directly comparable with the 0.4 ms per transaction ~~), comparable to GAT and well within requirements for real-time fraud detection systems. Breakdown:~~

- ~~• HRAPE encoding: 8ms~~
- ~~• SEAL layers ($\times 3$): 35ms~~
- ~~• Dual-head prediction: 3ms~~
- ~~• Mask generation (explanation): 4ms~~

~~Post-hoc explainers like GNNExplainer require 50–200ms per explanation (Ying et al., 2019), making them prohibitive for real-time deployment. RTXGNN generates explanations as byproduct of inference with negligible overhead (4ms), enabling production deployment at scale reported for SEFraud (Li et al., 2024), which refers to GPU inference amortised over large batches. Under our protocol, SEFraud (1.7 ms for one transaction) is somewhat faster than RTXGNN, since it has no node masks and no temporal encoding.~~

5.11. Case studies

~~We analyze impact of hidden dimension size on model performance. Figure ?? shows performance improves with model capacity up to hidden dimension of 128, after which it plateaus.~~

~~Table 16 shows the explanations of four test transactions: two correctly detected illicit transactions and one correctly classified licit transaction from step 42, and one illicit transaction from step 44 that RTXGNN misses. Because Elliptic’s features are anonymised, we report feature indices together with their documented group (local or aggregated) and do not attach any further meaning to them.~~

~~We select $d = 128$ as optimal trade-off between accuracy and computational efficiency. Smaller dimensions ($d = 32, 64$) underfit the 166-dimensional feature space. Larger dimensions ($d = 256, 512$) provide negligible improvement ($+0.3\%$ F1) while doubling training time. The plateau at $d = 128$ suggests this capacity sufficiently captures fraud patterns in the Elliptic dataset. Three observations follow from these cases and from the statistics of the masks over all test transactions. First, the feature mask is nearly binary. For a typical test~~

Table 16: Explanations of four Elliptic test transactions by RTXGNN (seed 0). Open features: number of features with mask value above 0.5 (in parentheses: local ones). The first five open features are listed in the order of the mask generator’s pre-sigmoid score; groups: L = local, A = aggregated (the only semantics documented for the anonymised features). Score in parentheses: after removing these five features. Neighbours: the most important incoming and outgoing transactions (edge importance I_{uv}) with their labels.

Case	score (w/o top-5)	open fea- tures (lo- cal)	first five open features: index (group)	top neighbours: direction, label, importance
true positive, step 42	1.00 (0.99)	82 (44)	162 (A), 28 (L), 160 (A), 89 (L), 84 (L)	out., unlabelled, 0.00; inc., licit, 0.00
true positive, step 42	1.00 (1.00)	79 (41)	28 (L), 162 (A), 160 (A), 89 (L), 84 (L)	out., unlabelled, 0.23; inc., illicit, 0.02
true negative, step 42	0.01 (0.02)	75 (41)	162 (A), 160 (A), 88 (L), 28 (L), 120 (A)	out., unlabelled, 0.00; out., unlabelled, 0.00; inc., licit, 0.00
false negative (after step 43), step 44	0.01 (0.01)	77 (44)	120 (A), 141 (A), 160 (A), 88 (L), 28 (L)	out., unlabelled, 0.00

transaction, about 30% of the features are fully open ($\text{gate} > 0.99$), about 30% fully closed (< 0.01), and 75–82 features are open in the four cases, roughly half of them local. The mask therefore acts as a per-transaction feature selection rather than as a fine-grained ranking. The ordering within the open set in Table 16 is by the mask generator’s pre-sigmoid score. Second, the leading features are largely shared across transactions, i.e. the selection is mostly global. Third, removing the first five open features changes the scores by at most 0.005, in line with the low necessity found in Section 5.7. The missed transaction after step 43 receives a confident licit score, and its explanation offers no warning sign. This illustrates that explanations do not help with new patterns that the model has never seen. For decision support, the most useful elements are thus (i) the retained feature set, which documents which inputs could have influenced the score, (ii) the neighbouring transactions with non-negligible importance, which point the analyst to related cases, and (iii) the score itself, used to rank the review queue.

6. Discussion

~~Our comprehensive evaluation demonstrates RTXGNN achieves state-of-the-art fraud detection performance (~~

6.1. *What the detection results mean for decision making*

Under a common protocol, RTXGNN is among the most accurate graph models on Elliptic, and gradient-boosted trees are more accurate still. The differences between the best GNNs are of the same order as the variation across seeds. An F1 =0.5129, AUC=0.8661) while maintaining intrinsic interpretability across multiple evaluation dimensions: of about 0.7 means, at the validation-chosen threshold, that about a quarter of the illicit test transactions are missed and that about a third of the alerts are false (Section 5.2). No financial institution would run such a model as an autonomous decision maker. The realistic use is as a prioritisation tool in a human review queue, where the operating point is set by the capacity of the investigation team (Table 4) and the model is combined with rules and other controls. In this role, three properties matter beyond F1: ranking quality at the top of the list (AP), the effort needed to review an alert, and robustness over time. The contribution of RTXGNN is mainly the second one. It provides a sufficient and stable attribution with every score at a small additional inference cost (0.24 ms per single request), while its detection performance is on par with the strongest GNNs.

6.2. *Explanations, regulation and how masks could be operationalised*

The masks do not by themselves make a system compliant with GDPR Article 22 or the ECOA. The regulations cited in Section 2 impose obligations on institutions and processes: lawful basis and human involvement for automated decisions, notices with specific reasons, documented and validated models, and auditable AML decisions. No model architecture can satisfy these obligations by itself. Several of them also do not concern the Elliptic setting at all. The ECOA, for instance, governs consumer credit decisions, not the screening of cryptocurrency transactions. A ranking of anonymous feature indices, as produced on Elliptic, is not a meaningful explanation for a data subject, an analyst or a regulator.

The masks can nevertheless support such processes when the institution controls the feature definitions. We see four concrete requirements.

1. ~~: +4.2% F1 over best explainable baseline (SEFraud), +5.6% over best fraud-specific baseline (PC-GNN), +11.2% over best temporal baseline (DySAT)~~

2. ~~∴ HRAPE contributes +2.53pp F1, sparsity regularization +6.47pp (largest contribution), SEAL +4.17pp, confirming all components essential~~
3. ~~∴ Focal loss, class weighting, and sparsity-driven feature selection achieve strong precision-recall balance (P=0.4423, R=0.6106)~~
4. ~~∴ Maintains 76% of full-supervision performance with only 10% labels, enabling practical deployment in label-scarce scenarios~~
5. ~~∴ Strong near-term performance (F1 $\approx$ 0.5–0.7 for steps 35–41), though vulnerable to abrupt concept drift requiring retraining~~ **A governed feature dictionary.** Every model input needs a documented, business-level definition, for example “number of distinct counterparties in the last 7 days”. Features derived from the graph also need a description of how they are computed. Top- k feature attributions can then be translated into candidate reason statements. This is impossible on Elliptic, whose features are anonymised, and is the main reason we only report feature *groups* (local versus aggregated) in the case studies.
6. ~~∴ Positive fidelity score (0.0764) validates SEAL identifies genuine prediction drivers~~ **Reason codes curated by experts.** Attributions indicate which inputs the score depended on and in which direction the prediction moves when they are removed. They do not justify the decision. A curated mapping from (groups of) features to reason codes, reviewed by compliance officers, and a rule for which reasons are reported (for example the top reasons that push the score towards “illicit”) are needed before any attribution reaches a customer notice or a suspicious-activity report.
7. ~~∴ Competitive performance on Amazon Computers (F1=0.7234), perfect detection on synthetic fraud rings (F1=1.0000)~~ **Human review and audit trail.** The score, the threshold version, the model version, and the feature and edge attributions of every alert can be logged without a separate explanation step, because they are produced by the forward pass that produced the score. Such a log supports the documentation expected in AML investigations and in model-risk management (Board of Governors of the Fed). The final decision remains with the analyst.
8. ~~∴ Sub-50ms inference latency with real-time explanation generation, 1000 $\times$ faster than post-hoc explainers~~ **Validation of the explanations themselves.** Before explanations are used in any customer-facing or supervisory context, their faithfulness, their stability and their usefulness to analysts must be validated. We report faithfulness and stability for

RTXGNN (Section 5.7). Its explanations are stable under small input perturbations but agree only weakly across independently retrained models, so explanations must be versioned together with the model that produced them. Usefulness to analysts requires a user study with practising analysts, which we have not conducted.

These results position RTXGNN as a practical solution for production fraud detection requiring both accuracy and regulatory-compliant explainability. The framework bridges the gap between black-box GNN performance and transparent decision-making demanded by financial compliance environments.

RTXGNN’s intrinsic explainability mechanism directly addresses the transparency requirements imposed on automated financial decision-making systems. GDPR Article 22 prohibits solely automated decisions with significant effects unless accompanied by meaningful information about the underlying logic. SEAL’s multi-granularity explanations—covering feature, node, edge, and temporal importance—satisfy this requirement without post-hoc overhead. While the academic debate over whether GDPR creates an enforceable right to explanation remains unresolved (Wachter et al., 2017; Selbst and Powles, 2017; Sargeant, 2026), regulatory practice increasingly demands transparency regardless of legal interpretation. Under these conditions, the masks are best regarded as evidence that can support transparency obligations. They do not demonstrate compliance with GDPR, ECOA, FATF guidance, MiCA or the EU AI Act (European Parliament and Council of

In the United States, the Equal Credit Opportunity Act (U.S. Congress, 1974) and the Consumer Financial Protection Bureau’s 2022 circular explicitly prohibit the use of complex algorithms when doing so prevents providing specific adverse action reasons—a direct challenge to black-box deployments. RTXGNN addresses this by surfacing ranked feature identifiers (e.g., Features 51, 54, 52 for Transaction 145713) that can be mapped to business-interpretable descriptions maintained by the institution. Similar transparency expectations apply under FATF guidance⁷ and the MiCA regulation⁷, where automated flagging of suspicious cryptocurrency transactions must be auditable.

An important practical gap remains, however: machine-generated feature importance scores are not self-interpreting. Financial institutions must maintain

⁷
⁷

mapping tables from model feature indices to domain-meaningful attributes before RTXGNN’s outputs fully satisfy regulatory scrutiny. Bridging this gap requires organizational investment alongside the technical solution.

The dominant assumption in explainable AI is that interpretability constrains model capacity, creating an inherent accuracy trade-off. RTXGNN’s ablation results challenge this assumption. The sparsity loss—whose primary purpose is to produce concise, human-readable explanations—delivers the largest single performance gain among all components. This occurs because sparse masks force the model to commit to a small set of discriminative signals rather than exploiting distributed, low-confidence correlations that generalize poorly. In a severely imbalanced setting, this selective pressure mitigates overfitting to the majority class without requiring additional architectural complexity.

6.3. Mask sparsity is regularisation, not a new kind of regularisation

This reframes explainability not as a constraint imposed on a trained model, but as a training-time inductive bias that improves the learned representation. Post-hoc methods such as GNNExplainer (Ying et al., 2019) and SubgraphX (Yuan et al., 2021) cannot replicate this effect because they optimize explanations after the model has already converged; the model’s internal representations remain unaffected. RTXGNN demonstrates that incorporating interpretability objectives during training yields benefits post-hoc methods structurally cannot provide. The sparsity term is an L_1 penalty on auxiliary gate activations, and a claim about its regularising effect must be tested against standard regularisers. Our controlled comparison (Section 5.4) does not show a large effect. With 5 seeds, the differences between mask sparsity, stronger dropout and stronger weight decay are within the seed-to-seed variation. We therefore make only the weaker claim that the explanation objective does not cost detection performance. We do not claim that it improves it.

6.4. Deployment: drift, retraining and latency

The temporal stability results reveal a practical boundary for RTXGNN’s deployment. The model generalizes well to gradual distributional change but fails under abrupt regime shifts—a limitation shared by all static-trained models. In production, this implies that monitoring systems must track per-step performance and trigger retraining before degradation compounds. The label efficiency results are directly relevant here: retraining on as little

as 10% of new labeled transactions recovers the majority of full-supervision performance, making frequent update cycles operationally feasible.

At institutional scale, GPU-batched inference sustains sub-millisecond per-transaction latency, supporting real-time deployment. Following the Lambda architecture pattern of systems such as BRIGHT (Lu et al., 2022), pre-computing embeddings for stable entities in batch pipelines while reserving online inference for new transactions reduces computational load without sacrificing explanation capability. RTXGNN is best positioned as a complement to existing rule-based systems rather than a replacement: graph-based reasoning captures relational and temporal fraud signals that rules struggle to encode, while rules provide deterministic coverage for known typologies. All models collapse on Elliptic after time step 43. This is not specific to graph models or to RTXGNN. Weber et al. (2019) attribute it to the shutdown of a large dark market, after which the illicit activity in the data changes. The retraining experiment (Section 5.5) shows that refitting with a one-step label delay recovers part of the lost performance for both RTXGNN and the random forest. It cannot anticipate a new illicit pattern before labels for it exist. A deployment would therefore need drift monitoring on the score distribution and on the alert yield, periodic retraining with a documented schedule, and fallback rules for known typologies. Our latency measurements (Section 5.10) show that the model itself fits comfortably in an online budget on a single CPU core. In practice the budget is dominated by the retrieval of the transaction neighbourhood from the institution’s data store, which we emulate with an in-memory graph. Latency in a production system will also depend on that store.

6.5. Limitations

Three limitations warrant explicit acknowledgment. First, evaluation is primarily conducted on the Elliptic Bitcoin dataset, whose characteristics—pseudonymous addresses, public blockchain, UTXO transaction model—differ substantially from card-present fraud, ACH transfers, or Ethereum-based decentralized finance. Generalizability to these settings requires dedicated validation.

Second, explanation quality is assessed through automated fidelity metrics and illustrative case studies

- **Data.** Elliptic’s snapshots contain no cross-snapshot edges and no intra-snapshot timestamps, so the temporal components of RTXGNN are close to inactive on it. T-Finance has no timestamps at all. The

evidence for HRAPE therefore rests on its comparison with Time2Vec on Elliptic and on synthetic data. Validation on a real transaction dataset with continuous timestamps and repeated account activity is the most important next step.

- **Baselines.** All graph baselines are our re-implementations under one protocol and budget, without per-model tuning, and some were adapted from multi-relation or continuous-time settings. They may perform better with their original code and tuning.
- **Explanations.** Faithfulness is measured with perturbation metrics that replace features by the training mean. Such perturbations can move inputs off the data manifold. Our training objective also uses perturbations of this kind, which may favour RTXGNN on these metrics. We mitigate this by applying the post-hoc explainers to the same trained model and by reporting a random-explanation reference. The feature masks are nearly binary and weakly necessary, so they identify a set of features the prediction may depend on rather than a ranked list of indispensable ones. No user study ~~with practicing fraud analysts has been conducted to verify that SEAL’s outputs meaningfully support investigative decisions or satisfy regulators in practice. Cherif et al.’s systematic review (Cherif et al., 2024) identifies this human evaluation gap as field-wide; addressing it is a necessary step before making strong compliance claims.~~

~~Third, scalability to billion-node institutional graphs remains undemonstrated. The current architecture requires graph partitioning, distributed training infrastructure, and approximate neighborhood sampling strategies not evaluated here.~~ has been carried out.

- **Compute.** All experiments were run on a 4-vCPU machine and a single T4 GPU. Training cost is about three times that of the same network without the explanation objective. Scalability to graphs with billions of edges was not evaluated.

~~Future work should prioritize: human-centered evaluation with domain experts; adversarial robustness against feature manipulation by adaptive fraudsters; continual learning mechanisms for handling abrupt concept drift without full retraining; and federated variants enabling cross-institutional collaboration under data privacy constraints.~~

7. Conclusions

~~This paper introduced We presented~~ RTXGNN, a self-explainable self-explaining temporal graph neural network framework for real-time financial fraud detection that treats interpretability as a first-class design requirement rather than a post-hoc addition. The central finding is that intrinsic explainability and predictive accuracy are not in tension: when incorporated during training, sparsity-driven interpretability constraints act as domain-informed regularization that benefits both objectives simultaneously.

Three limitations bound the current results and define the agenda for future work. First, evaluation is concentrated on the EllipticBitcoin dataset; validation on card-present fraud, ACH transfers, and decentralized finance settings is necessary before broad deployment claims can be made. Second, no user study with practicing fraud analysts has been conducted, leaving open the question of whether SEAL’s outputs meaningfully support investigative decisions in practice. Third, for fraud detection. Its masks gate message passing and are returned as the framework is vulnerable to abrupt concept drift, as demonstrated by the sharp performance collapse explanation of every score, and its training objective makes the top- k explanation sufficient and its complement uninformative. The temporal encoding (HRAPE) uses only relative time. In a controlled comparison with 16 baselines over 10 seeds on Elliptic, RTXGNN matches the best graph neural network (F1 0.700, AP 0.771) and is ahead of the self-explaining SEFraud on average. Its ranking is precise at the top of the alert queue, which is where analysts spend their effort. Gradient-boosted trees on Elliptic’s engineered features remain more accurate (F1 0.790). They do not, however, produce explanations over the relations between transactions. RTXGNN’s explanations come with every score, without a separate explainer and at a small cost. They are nearly as sufficient as Integrated Gradients and much more stable than GNNExplainer, and on synthetic data they single out the planted laundering transactions. A single transaction is scored and explained end to end in 5.9 ms (99th percentile) on one CPU core.

The evaluation also shows the limits of the approach. All models fail after the regime change at time step 42; ~~continual learning mechanisms or online retraining protocols are required for sustained production robustness~~⁴³, and only retraining on newly labelled data recovers part of the performance. The explanations are weakly necessary: removing the top features rarely changes the prediction, because Elliptic’s features are redundant. HRAPE carries

little information on Elliptic’s snapshots and is unavailable on T-Finance, so its benefit rests mainly on comparisons with Time2Vec and on synthetic evidence. Finally, the explanations support, but do not by themselves establish, the transparency expected by regulation. That requires governed feature definitions, curated reason codes, human review and validation with analysts.

~~Promising directions for extending this work include adaptation to continuous-time dynamic graphs, integration of causal explanation frameworks to move beyond associative masks, federated variants enabling privacy-preserving collaboration across institutions, and adversarial robustness evaluation against adaptive fraudsters who may exploit known explanation mechanisms.~~ Future work should (i) evaluate the temporal components on real transaction data with continuous timestamps and repeated account activity, (ii) study continual-learning and drift-detection strategies that shorten the recovery after regime changes, (iii) conduct a user study with fraud analysts on the usefulness of the explanations, and (iv) assess the robustness of models and explanations against adversaries who know the explanation mechanism.

~~The implementation of RTXGNN is publicly~~

Data and code availability

The implementation, the experiment scripts, the per-run result records and the scripts that generate every table and figure of this paper are available at https://github.com/vinhq dang/explainable_GNN_financial_fraud ~~to facilitate reproducibility and future research.~~ The Elliptic dataset is available from its publishers (Weber et al., 2019). T-Finance is distributed with GADBench (Tang et al., 2023). The synthetic benchmark is generated by `rtxgnn/synthetic.py`.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT in order to polish the writing. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Appendix A. Elliptic: class counts per time step

Appendix B. Adaptation of the baselines

Table A.17: Labelled and unlabelled transactions per time step of Elliptic and the split each step belongs to.

Step	Split	Licit	Illicit	Unlab.	Step	Split	Licit	Illicit	Unlab.
1	train	2,130	17	5,733	26	train	421	96	2,006
2	train	1,099	18	3,427	27	train	182	24	883
3	train	1,268	11	5,342	28	train	199	85	1,369
4	train	1,410	30	4,253	29	train	845	329	3,101
5	train	1,874	8	4,921	30	train	441	83	1,959
6	train	480	5	3,843	31	val	604	106	2,106
7	train	1,101	102	4,845	32	val	981	342	3,202
8	train	1,098	67	3,292	33	val	418	23	2,710
9	train	530	248	4,218	34	val	478	37	1,971
10	train	954	18	5,755	35	test	1,159	182	4,166
11	train	565	131	3,600	36	test	1,675	33	4,685
12	train	490	16	1,541	37	test	458	40	2,808
13	train	518	291	3,719	38	test	645	111	2,135
14	train	374	43	1,605	39	test	1,102	81	1,577
15	train	471	147	3,021	40	test	1,099	112	3,270
16	train	402	128	2,445	41	test	1,016	116	4,210
17	train	712	99	2,574	42	test	1,915	239	4,986
18	train	337	52	1,587	43	test	1,346	24	3,693
19	train	665	80	2,761	44	test	1,567	24	3,384
20	train	640	260	3,391	45	test	1,216	5	4,377
21	train	541	100	2,896	46	test	710	2	2,807
22	train	1,605	158	4,131	47	test	824	22	4,275
23	train	1,134	53	2,978	48	test	435	36	2,483
24	train	989	137	3,466	49	test	420	56	1,978
25	train	476	118	1,720					
Total		42,019	4,545	157,205					

All baselines use the shared protocol of Section 4.4 (hidden size 64, two layers, dropout 0.2, class-balanced loss, Adam, early stopping on validation AP, threshold tuned on validation). The code is in `rtxgnn/baselines.py`.

- **GCN, GraphSAGE, GAT.** PyTorch Geometric layers on the symmetrised payment graph; GAT uses 4 heads.
- **EvolveGCN-O.** The weight matrix of each of the two GCN layers is the hidden state of an LSTM whose input is the previous weight matrix (rows are treated as the batch). The LSTM steps once per time step over all 49 snapshots, including validation and test snapshots, which

are processed without labels. Each snapshot is convolved with its own normalised adjacency.

- **TGAT.** Two temporal attention layers. Keys and values are built from $[\mathbf{h}_u \parallel \Phi(\Delta_{uv}) \parallel r_{uv}]$ and queries from $[\mathbf{h}_v \parallel \Phi(0)]$, with the functional time encoding $\Phi(\Delta) = \cos(\omega\Delta + \mathbf{b})$ (Xu et al., 2020). All neighbours are used, since Elliptic degrees are small.
- **TGN.** A GRU memory is updated from messages $[\mathbf{s}_v \parallel \mathbf{s}_u \parallel \Phi(t - t_v^-) \parallel \mathbf{x}_u]$ aggregated by their mean. The embedding module is one TGAT layer over $\text{ReLU}(\mathbf{W}\mathbf{x} + \mathbf{W}'\mathbf{s})$. Events are processed in chronological batches: one batch per time step on Elliptic, and one batch per day on the synthetic data. Embeddings of the nodes involved in a batch are computed from the memory after the batch update. On Elliptic every node takes part in the events of a single time step only, so the memory before that step is always empty.
- **APAN.** Every node has a mailbox of its $K = 10$ most recent mails. Its embedding is multi-head attention of its own state over the mailbox followed by an MLP. Mails $\mathbf{W}[\mathbf{z}_u \parallel \mathbf{z}_v \parallel \Phi(\Delta_{uv})]$ are propagated to the 1-hop neighbours. On snapshot data we run one encoding pass, one propagation round and a second encoding pass per step, instead of the asynchronous propagation of the original system (Wang et al., 2021).
- **CARE-GNN.** Single-relation version: a label-aware similarity network (a linear classifier per layer, trained with an auxiliary cross-entropy of weight 2) defines the distance $|p_u - p_v|$. Each node keeps the top- p fraction of its neighbours with the smallest distance, and p is adapted after every epoch by ± 0.02 when the average distance of the selected neighbours decreases or increases (the reinforcement-learning rule of Dou et al., 2020). Aggregation is $\text{ReLU}(\mathbf{W}[\mathbf{h}_v \parallel \text{mean}_{u \in \text{sel}} \mathbf{h}_u])$.
- **PC-GNN.** Pick: in every epoch the loss is computed on a label-balanced sample of training nodes, drawn with probability proportional to $(\deg(v) + 1)/|\text{class}(v)|$, with four times as many nodes as there are illicit training nodes. Choose: neighbours whose label-aware distance exceeds $\rho = 0.5$ are dropped, and every illicit training node receives two additional neighbours drawn from the illicit training nodes. Label-aware predictor with auxiliary loss weight 2.

- **GAS.** Two GCN branches, one on the payment graph and one on a k -nearest-neighbour graph ($k = 5$, cosine similarity of features) built within each time step, concatenated before the classifier. This replaces the review-content similarity graph of the original.
- **FRAUDRE.** Layers $\text{ReLU}(\mathbf{W}[\mathbf{h}_v \parallel \text{mean}_u \mathbf{h}_u \parallel \text{mean}_u \mathbf{h}_u - \mathbf{h}_v])$ (fraud-aware convolution with node-neighbour differences). The classifier takes the concatenation of the input and of all layer outputs, and imbalance is handled by the class-balanced loss.
- **SEFraud.** A feature-mask MLP and an edge-mask MLP gate a two-layer graph transformer, as in Li et al. (2024) but on a single node and edge type. Its loss adds to the classification loss a triplet loss (margin 1, weight 0.5) with anchor the embedding of the unmasked graph, positive the embedding of the masked graph and negative the embedding of the complementary graph, plus an L_1 mask penalty of weight 0.01.

Appendix C. Effect of feature preprocessing

Table C.18: Test results on Elliptic (seed 0) with raw features versus quantile-normalised features (Section 4). All other settings are identical (the raw-feature runs were trained on the GPU machine). Validation AP is computed on steps 31–34, which are used for model selection.

Model	raw features			quantile-normalised		
	val. AP	test F1	test AP	val. AP	test F1	test AP
MLP	0.839	0.553	0.500	0.930	0.625	0.745
GCN	0.872	0.527	0.441	0.964	0.692	0.726
GraphSAGE	0.939	0.537	0.508	0.977	0.711	0.766
GAT	0.900	0.544	0.539	0.963	0.721	0.748
SEFraud	0.959	0.514	0.487	0.957	0.651	0.729
RTXGNN (ours)	0.950	0.530	0.522	0.964	0.664	0.759

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